

Supplementary materials for the paper “Tensor Time Series Imputation through Tensor Factor Modelling” - Extra simulations, NYC taxi data analysis, and proof of all theorems and propositions

1 Extra simulation results and data analysis

1.1 Sensitivity for the tuning parameter in Theorem 5

We provide some simulation results on the performance of our number of factor estimator $\hat{r}_k$ relative to the choice of ξ in Theorem 5. For demonstration purpose, we adapt the general setup depicted in Section 5.1 to generate the loading matrices, factor and noise series, except that only $\mathcal{N}(0,1)$ is used to generate the innovation process. On the dimension of data, we consider order $K = 1, 2, 3$ such that

- $K = 1$: $T = 160, d_1 = 80$;
- $K = 2$: $T = d_1 = d_2 = 40$;
- $K = 3$: $T = d_1 = d_2 = d_3 = 20$;

where we assume two pervasive factors on each mode, i.e. true $r_k = 2$ and $\zeta_{k,j} = 0$ for all k, j .

To show the robustness of our choice of ξ , each setting is repeated 1000 times under four missing patterns: fully observed, (M-i), (M-ii) and (M-iii). See Section 5.1 for the description of these missing patterns.

Correct Proportion of $\hat{r}_1 = r_1 = 2$					
Missing Pattern	$\xi = 0.002$	$\xi = 0.02$	$\xi = 0.2$	$\xi = 2$	$\xi = 20$
$K = 1$ ($T = 160, d_1 = 80$)					
Fully observed	.999	.999	1	.986	.909
(M-i)	.999	.999	1	.985	.906
(M-ii)	.997	.997	.995	.985	.903
(M-iii)	.99	.989	.987	.934	.791
$K = 2$ ($T = d_1 = d_2 = 40$)					
Fully observed	1	1	1	.992	.843
(M-i)	1	1	1	.993	.842
(M-ii)	1	1	1	.986	.842
(M-iii)	.994	.995	.996	.972	.815
$K = 3$ ($T = d_1 = d_2 = d_3 = 20$)					
Fully observed	.999	.999	.997	.967	.732
(M-i)	.999	.999	.997	.967	.73
(M-ii)	.998	.998	.996	.959	.718
(M-iii)	.995	.997	.995	.95	.695

Table 1: Results of correct proportion for the number of factor estimator on mode-1 in 1000 replications.

As the dimension and the number of factors along each tensor mode is the same (within any setting), it suffices to study the correct proportion of $\hat{r}_1 = r_1 = 2$. The result for different values of

$\xi \in \{0.002, 0.02, 0.2, 2, 20\}$ is shown in Table 1. It is clear from the results that relatively small values of ξ should help to estimate the number of factors consistently. In particular, ξ ranges from 0.02 to 0.2 should work sufficiently well.

1.2 Introduction of new measure

We propose a new measure to gauge the imputation performance in this section. Let the observed data be $\mathbf{y} = (y_1, y_2, \dots, y_N)'$ and a vector of imputed data be $\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_N)'$. WLOG, we let the entries in $\mathbf{y}$ be ordered such that $y_1 \leq y_2 \leq \dots \leq y_N$, and each $\hat{y}_i$ is the corresponding imputed value for y_i . Given $q \in [N]$ and any integer $j = 0, 1, \dots, q$, denote by $q_j \in [N]$ the index of entry with value closest to the (j/q) -th quantile of $\mathbf{y}$. Hereafter, quantiles are always with respect to $\mathbf{y}$ when referred to. For $n \in [q]$, define

$$\mu_n := \frac{1}{q} \sum_{i=q_{n-1}}^{q_n} y_i, \quad \hat{\mu}_n := \frac{1}{q} \sum_{i=q_{n-1}}^{q_n} \hat{y}_i.$$

Hence, μ_n is the average of the entries in $\mathbf{y}$ between the $((n-1)/q)$ -th quantile and the (n/q) -th quantile.

With these, we define the *q-quantile relative squared error* (*q-RSE*) of $\hat{\mathbf{y}}$ on $\mathbf{y}$ as

$$q\text{-RSE} := \sum_{n=1}^q (\mu_n - \hat{\mu}_n)^2 \bigg/ \sum_{n=1}^q \mu_n^2 = \sum_{n=1}^q \left\{ \sum_{i=q_n}^{q_{n+1}} (y_i - \hat{y}_i) \right\}^2 \bigg/ \sum_{n=1}^q \left\{ \sum_{i=q_n}^{q_{n+1}} y_i \right\}^2.$$

It is obvious that *q-RSE* is always non-negative, and *q-RSE* = 0 if $\hat{y}_i = y_i$ for all $i \in [N]$. Note that the relative MSE defined in (5.1) is equivalent to *q-RSE* with $q = N$, and essentially the average within each quantile interval smooths out the idiosyncratic noise in each data point. Intuitively, *q-RSE* can be considered as a measure of how good the imputation of y_i is, compared with the overall level of y_i in each quantile interval.

As an example to demonstrate the pitfall of relative MSE, consider a one-factor model for vector time series and y_i represents some data entry for $i \in [N]$. Suppose $y_i = c_i + \epsilon_i$ and $\hat{y}_i = c_i$, where c_i is the common component to be imputed and ϵ_i is the noise. Hence we have perfect imputation here, and a good measure should have value close to 0. However,

$$\text{relative MSE} = \frac{\sum_{i=1}^N \epsilon_i^2}{\sum_{i=1}^N (c_i + \epsilon_i)^2},$$

which might even be larger than 1 if the noise term is comparable to the common component. For simplicity if we set $q = 1$, then

$$q\text{-RSE} = \frac{(\sum_{i=1}^N \epsilon_i)^2}{(\sum_{i=1}^N c_i + \sum_{i=1}^N \epsilon_i)^2},$$

which goes to 0 if we assume sensibly that as N gets large, $\sum_{i=1}^N \epsilon_i$ goes to 0 at a faster rate than $\sum_{i=1}^N c_i$. Thus, given that we cannot distinguish between the common component and the noise term for a given set of data, the *q-RSE* is a better measure to gauge imputation performances.

1.3 NYC taxi traffic

We analyze a set of taxi traffic data in New York city in this example. The data includes all individual taxi rides operated by Yellow Taxi in New York City, published at

<https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.

For simplicity, we only consider the rides within Manhattan Island, which comprises most of the data. The dataset contains 842 million trip records within the period of January 1, 2013 to December 31, 2022. Each trip record includes features such as pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts. Our example here focuses on the pick-up and drop-off dates/times, and pick-up and drop-off locations.

To classify the pick-up and drop-off locations in Manhattan, they are coded according to 69 predefined zones in the dataset. Moreover, each day is divided into 24 hourly periods to represent the pick-up and drop-off times each day, with the first hourly period from 0 a.m. to 1 a.m. The total number of rides moving among the zones within each hour are recorded, yielding data $\mathcal{Y}_t \in \mathbb{R}^{69 \times 69 \times 24}$ each day, where $y_{i_1, i_2, i_3, t}$ is the number of trips from zone i_1 to zone i_2 and the pick-up time is within the i_3 -th hourly period on day t . We consider business days and non-business days separately, so that we will analyze two tensor time series. The business-day series is 2,519 days long, while the non-business-day series is 1,133 days long, within the period of January 1, 2013 to December 31, 2022.

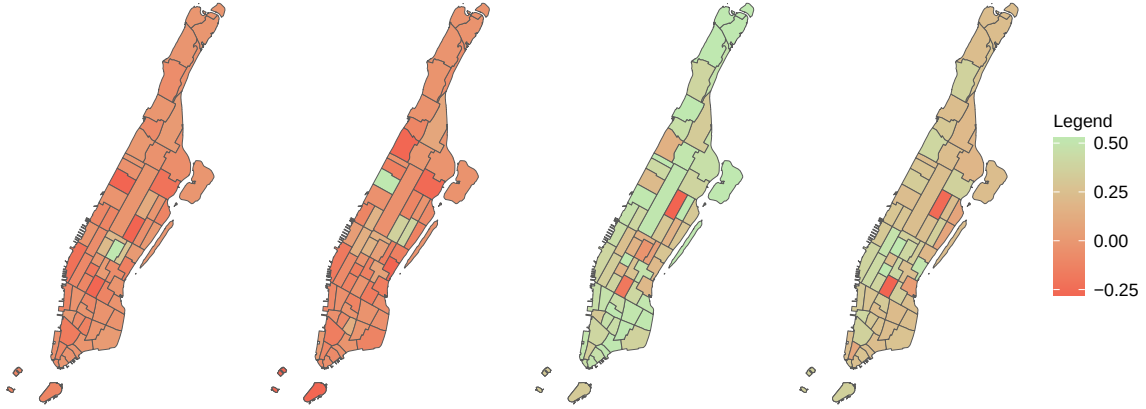


Figure 1: Estimated loading matrix $\hat{\mathbf{Q}}_1$ on four pick-up factors for business-day series. The data is block missing according to (M-iii).

As we are more interested in estimating the tensor factor structure with missingness and hence imputation, we demean the data before imposing different missing patterns, and investigate their effects. See descriptions of missing pattern (M-i) to (M-iii) in Section 5.1. We first estimate the rank of the core tensor using our proposed eigenvalue ratio estimator. Inspired by the simulation results in Section 5.1.2, one approach is to obtain the initial rank estimates $\hat{r}_k$ for $k \in [K]$, and re-estimate the number of factors based on the imputation using $\hat{r}_k + 1$ as the number of factors along mode- k .

It appears that the re-estimated rank is (1, 1, 1) for both the business-day and non-business-day series, and both are consistent under missing patterns (M-i), (M-ii), (M-iii) and no missing. This also coincides with iTIP-ER of Han et al. (2022). However, BCorTh of Chen and Lam (2024) estimates (4, 2, 2) for

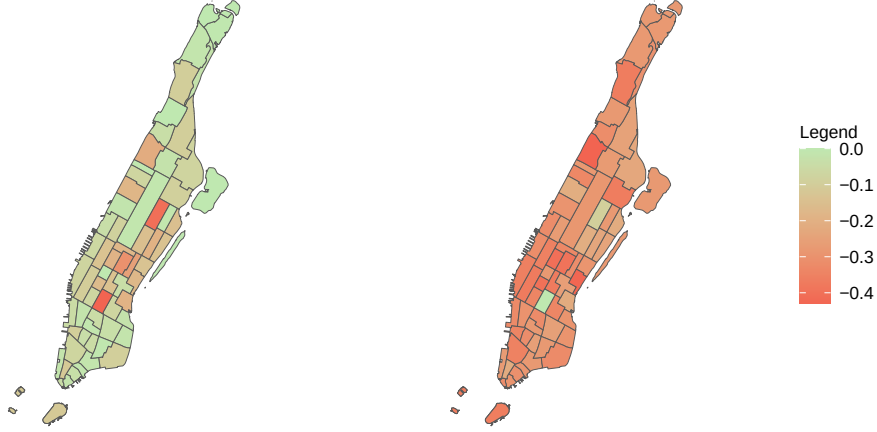


Figure 2: Estimated loading matrix $\hat{\mathbf{Q}}_2$ on two drop-off factors for business-day series. The data is block missing according to (M-iii).

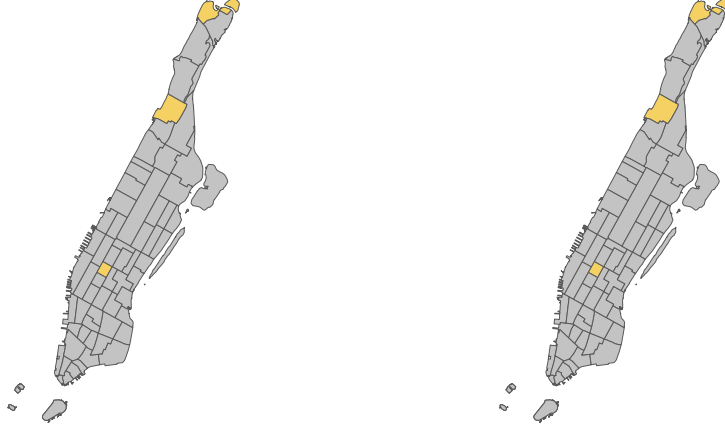


Figure 3: Testing zeros of $\mathbf{Q}_2$ on two drop-off factors for business-day series. Rows rejecting the null with 5% significance level are in grey, otherwise in yellow.

business-day series and $(3, 2, 2)$ for non-business-day series on fully observed data. Such discrepancy is due to the existence of weak factors in the NYC traffic data, and the rank $(1, 1, 1)$ is not sensible according to some exploratory studies in Chen et al. (2022). Although our factor number estimator has taken weak factors into account, the requirements for the factor strength are more stringent than the requirements when using BCorTh. To see this, consider the case of $K = 3$ and all factor strength are the same, then $\alpha_{k,j} = 2/3$ satisfies the rate assumption in Theorem 5.2 of Chen and Lam (2024) but fails our assumption in Theorem 5.

To cope with such weak factors, we use the rank $\hat{r}_k + r_*$ in the initial imputation for re-estimating

the rank, where r_* is a pre-specified integer characterising the maximum number of omitted factors. As discussed above, $r_* = 1$ might not be sufficient if weak factors are likely to exist. We use $r_* = 4$ here. As mentioned in Remark 2, we may re-estimate the rank using BCorTh of Chen et al. (2022) based on the imputed data. The resulting estimated ranks for (M-i), (M-ii) and (M-iii) align with those using full observations.

For convenience, we use the estimated rank $(\hat{r}_1, \hat{r}_2, \hat{r}_3) = (4, 2, 2)$ in the rest of the analysis to estimate the factor loading matrices for different modes. We note the resulting heatmaps of the estimated loadings are almost the same under different missing patterns, so we only display one particular example here. Figures 1 and 2 show the heatmaps of $\hat{\mathbf{Q}}_1$ and $\hat{\mathbf{Q}}_2$ for the business-day series under missing pattern (M-iii).

The two heatmaps reveal the same pattern in Manhattan, that traffic is mainly within the downtown areas (the upper east and around the Empire State Building), whereas recreation park or amusement areas (the Central Park and upper west) load slightly on the factor structure. In contrast, dining and sports areas (especially East Harlem) stand out in the factor loadings for non-business-day series. See Figures 4 and 5. In particular, the first diagram in Figure 2 is revealing that taxi drop-offs in business days are concentrated in the midtown center. To further find out active drop-off areas for business days, we statistically test if the loading entry is zero by leveraging Theorem 3 and Theorem 4. We test for each $i \in [69]$,

$$\mathcal{H}_0 : \mathbf{Q}_{2,i\cdot} = (0, 0)', \quad \mathcal{H}_1 : \text{otherwise.}$$

Test statistics obtained by $\|(\hat{\Sigma}_{HAC} + \hat{\Sigma}_{HAC}^\Delta)^{-1/2} \hat{\mathbf{D}}_2 \hat{\mathbf{Q}}_{2,i\cdot}\|^2$ follow a χ_3^2 distribution under the null. Using 5% significance level, the inactive zones (i.e., rows failing to reject the null) are highlighted in yellow, as presented in Figure 3. As we would expect, the common zones for the two factors represent parks and fashion production areas in general.



Figure 4: Estimated loading matrix $\hat{\mathbf{Q}}_1$ on four pick-up factors for non-business-day series. The data is block missing according to (M-iii).

The mode-3 loading matrix conveys the information of pick-up time, and the unveiled pattern is the

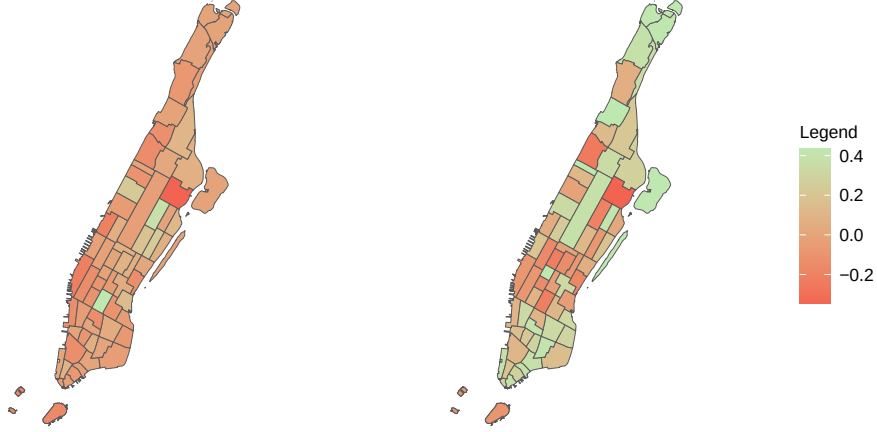


Figure 5: Estimated loading matrix $\hat{\mathbf{Q}}_2$ on two drop-off factors for non-business-day series. The data is block missing according to (M-iii).

same over different missing patterns. For better interpretation, we show $\hat{\mathbf{Q}}_3$ after a varimax rotation and scaling under missing pattern (M-iii) in Table 2 and Table 3 for the business-day and non-business-day series respectively. The highlighted parts indicate the rush-hours in business and non-business days, which depict drastically different patterns. From Table 2, we can see our first and second factors respectively capture the morning rush-hours (roughly from 9 a.m. to 10 a.m.) and the evening rush-hours (from 8 p.m. to 12 p.m.) for business days. Interestingly, we can see from the first factor that the taxi volume spikes at 4 p.m. which might be interpreted as an afternoon period. Table 3 gives similar patterns, except that all active hours are delayed. We may notice the night life in business days almost end after 1 a.m., while the traffic remains busy even after 2 a.m. in non-business days.

0 _{am}	12 _{pm}												12 _{am}											
	2	4	6	8	10	12 _{pm}	2	4	6	8	10	12 _{am}		2	4	6	8	10	12 _{am}					
1	2	1	0	0	0	-1	-3	-7	-10	-8	-7	-7	-9	-9	-9	-10	-8	-9	-8	-5	-2	-2	0	1
2	-9	-4	-2	-1	-1	0	2	5	4	0	-1	-1	2	0	1	1	-2	0	-3	-10	-13	-12	-13	-13

Table 2: Estimated loading matrix $\hat{\mathbf{Q}}_3$ for business-day series, after varimax rotation and scaling. The data is block missing according to (M-iii). Magnitudes larger than 9 are highlighted in red.

0 _{am}	2												4												6												8												10												12 _{pm}												2												4												6												8												10												12 _{am}																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
1	0	2	3	2	1	0	-1	-1	-4	-6	-9	-10	-11	-10	-10	-9	-8	-8	-8	-6	-3	-2	-2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0</

Table 3: Estimated loading matrix $\hat{\mathbf{Q}}_3$ for non-business-day series, after varimax rotation and scaling. The data is block missing according to (M-iii). Magnitudes larger than 9 are highlighted in red.

From the above discussion, it is easy to see that with our proposed imputation procedure, the tensor factor structure could be well studied even if we cannot observe the full data set. Nevertheless, Table 4 reveals that different missing patterns have a certain degree of influences on the imputation accuracy,

which is also suggested by previous simulation results. The q -RSE, introduced in Section 1.2, are reported for $q = 5, 10, 20, 100$, so that the overall performance is gauged accounting for large magnitudes of noise. Lastly, the analysis on the Manhattan Taxi data is for demonstration of our imputation method, and more sophisticated investigations are required to further understand the traffic pattern.

	business-day			non-business-day		
q	(M-i)	(M-ii)	(M-iii)	(M-i)	(M-ii)	(M-iii)
5	0.231	0.231	0.217	0.206	0.206	0.226
10	0.225	0.225	0.204	0.213	0.213	0.221
20	0.225	0.225	0.195	0.225	0.225	0.228
100	0.234	0.234	0.186	0.247	0.247	0.247

Table 4: q -RSE for business-day and non-business-day series under different missing patterns, with $q = 5, 10, 20, 100$.

2 Proof of all theorems and propositions

From Section 3.4, $\widehat{\mathbf{Q}}_k$ contains the eigenvectors corresponding to the first r_k largest eigenvalues of $\widehat{\mathbf{S}}_k$. Hence with $\widehat{\mathbf{D}}_k$ an $r_k \times r_k$ diagonal matrix containing all the eigenvalues of $\widehat{\mathbf{S}}_k$ (WLOG from the largest on the top-left element to the smallest on the bottom right element), we have $\widehat{\mathbf{S}}_k \widehat{\mathbf{Q}}_k = \widehat{\mathbf{Q}}_k \widehat{\mathbf{D}}_k$, so that

$$\widehat{\mathbf{Q}}_k = \widehat{\mathbf{S}}_k \widehat{\mathbf{Q}}_k \widehat{\mathbf{D}}_k^{-1}. \quad (\text{S.1})$$

To simplify notations, hereafter we fix k and only focus on the mode- k unfolded data. Define

$$\begin{aligned} \widehat{\mathbf{D}} &:= \widehat{\mathbf{D}}_k, \mathbf{Y}_t := \text{mat}_k(\mathcal{Y}_t), \widehat{\mathbf{S}} := \widehat{\mathbf{S}}_k, \psi_{ij,h} := \psi_{k,ij,h}, \mathbf{Q} := \mathbf{Q}_k, \\ \mathbf{\Lambda} &:= \mathbf{\Lambda}_k, \mathbf{F}_{Z,t} := \text{mat}_k(\mathcal{F}_{Z,t}), \mathbf{E}_t := \text{mat}_k(\mathcal{E}_t), \mathbf{H}_j := \mathbf{H}_{k,j}, \mathbf{H}^a := \mathbf{H}_k^a, \end{aligned} \quad (\text{S.2})$$

where $\mathbf{H}_{k,j}$ and $\mathbf{H}_k^a$ are defined in (4.3) and (4.4) respectively, and similarly to all respective hat versions of the above.

Before proving any theorems, we present and prove the following Proposition first.

Proposition 1 *Let Assumption (E1), (E2) and (F1) hold. Then*

1. *there exists a constant $c > 0$ so that for any $k \in [K], t \in [T], i_k \in [d_k]$ and $h \in [d_{-k}]$, we have*

$$\mathbb{E}\mathcal{E}_{t,i_1,\dots,i_K} = 0, \mathbb{E}\mathcal{E}_{t,i_1,\dots,i_K}^4 \leq c, \text{ and}$$

$$\begin{aligned} \sum_{j=1}^{d_k} \sum_{l=1}^{d_k} \left| \mathbb{E}[\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jl}] \right| &\leq c, \\ \sum_{l=1}^{d_k} \sum_{s \in \psi_{k,ij,l}} \left| \text{cov}(\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{il} \text{mat}_k(\mathcal{E}_s)_{jl}) \right| &\leq c; \end{aligned}$$

2. *there exists a constant $c > 0$ so that for any $k \in [K], i, j \in [d_k]$, and any deterministic vectors $\mathbf{u} \in \mathbb{R}^{r_k}$ and $\mathbf{v} \in \mathbb{R}^{r_{-k}}$ with constant magnitudes,*

$$\mathbb{E} \left(\frac{1}{d_{-k}^{1/2}} \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|^{1/2}} \sum_{t \in \psi_{k,ij,h}} \text{mat}_k(\mathcal{E}_t)_{jh} \mathbf{u}' \text{mat}_k(\mathcal{F}_t) \mathbf{v} \right)^2 \leq c;$$

3. *for any $k \in [K], i, j \in [d_k], h \in [d_{-k}]$,*

$$\frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} \text{mat}_k(\mathcal{F}_t) \text{mat}_k(\mathcal{F}_t)', \quad \frac{1}{T} \sum_{t=1}^T \text{mat}_k(\mathcal{F}_t) \text{mat}_k(\mathcal{F}_t)' \xrightarrow{p} \mathbf{\Sigma}_k := r_{-k} \mathbf{I}_{r_k},$$

with the number of factors r_k fixed as $\min\{T, d_1, \dots, d_K\} \rightarrow \infty$. For each $t \in [T]$, all elements in $\mathcal{F}_t$ are independent of each other, with mean 0 and unit variance.

Our consistency results in Theorem 1 can be proved assuming the three implied results from Proposition 1, on top of Assumption (O1), (M1), (L1) and (R1). Result 1 from Proposition 1 can be a stand alone assumption on the weak correlation of the noise $\mathcal{E}_t$ across different dimensions and times, while result 2 can be on the weak dependence between the factor $\mathcal{F}_t$ and the noise $\mathcal{E}_t$. Finally, result 3 can be a stand alone assumption on the factors $\mathcal{F}_t$.

Proof of Proposition 1. We have $\mathbb{E}(\mathcal{E}_t) = 0$ from Assumption (E1). Next we want to show that for any $k \in [K], t \in [T]$ and $i_k \in [d_k]$, $\mathbb{E}\mathcal{E}_{t,i_1,\dots,i_K}^4$ is bounded uniformly. From (4.2), each entry in $\mathcal{E}_t$ is a sum of

two parts: a linear combination of the elements in $\mathcal{F}_{e,t}$, and the corresponding entry in $\boldsymbol{\epsilon}_t$. By Assumption (E2), we have

$$\mathbb{E}[(\boldsymbol{\epsilon}_t)_{i_1, \dots, i_K}^4] = \mathbb{E}\left[\left(\sum_{q \geq 0} a_{\epsilon, q}(\mathcal{X}_{\epsilon, t-q})_{i_1, \dots, i_K}\right)^4\right] \leq \left(\sum_{q \geq 0} |a_{\epsilon, q}|\right)^4 \sup_t \mathbb{E}[(\mathcal{X}_{\epsilon, t})_{i_1, \dots, i_K}^4] \leq C,$$

where $C > 0$ is a generic constant. It holds similarly that $\mathbb{E}[(\mathcal{F}_{e,t})_{i_1, \dots, i_K}^4] \leq C$ uniformly on all indices. With this, defining $\mathbf{A}^{\circ m} := \mathbf{A} \circ \dots \circ \mathbf{A}$ (elementwise m -th power),

$$\begin{aligned} \|\mathbb{E}\mathcal{E}_t^{\circ 4}\|_{\max} &= \|\mathbb{E}[\text{mat}_k(\mathcal{E}_t)]^{\circ 4}\|_{\max} \leq 8(\|\mathbb{E}[\mathbf{A}_{e,k} \text{mat}_k(\mathcal{F}_{e,t}) \mathbf{A}'_{e,-k}]^{\circ 4}\|_{\max} + \|\boldsymbol{\Sigma}_{\epsilon}^{\circ 4}\|_{\max} \cdot \|\mathbb{E}\boldsymbol{\epsilon}_t^{\circ 4}\|_{\max}) \\ &\leq 8(\|\mathbf{A}_{e,k}\|_{\infty}^4 \cdot \|\mathbf{A}_{e,-k}\|_{\infty}^4 \cdot \|\mathbb{E}\mathcal{F}_{e,t}^{\circ 4}\|_{\max} + \|\boldsymbol{\Sigma}_{\epsilon}^{\circ 4}\|_{\max} \cdot \|\mathbb{E}\boldsymbol{\epsilon}_t^{\circ 4}\|_{\max}) \\ &= 8\left(\prod_{j=1}^K \|\mathbf{A}_{e,j}\|_{\infty}^4 \cdot \|\mathbb{E}\mathcal{F}_{e,t}^{\circ 4}\|_{\max} + \|\boldsymbol{\Sigma}_{\epsilon}^{\circ 4}\|_{\max} \cdot \|\mathbb{E}\boldsymbol{\epsilon}_t^{\circ 4}\|_{\max}\right) \leq C, \end{aligned}$$

where $C > 0$ is again a generic constant, and we used Assumption (E1) in the last line and the fact that r_j is a constant for $j \in [K]$. This is equivalent to $\mathbb{E}\mathcal{E}_{t,i_1, \dots, i_K}^4 \leq c$ for some constant c .

With (4.2) in Assumption (E1), we have

$$\text{mat}_k(\mathcal{E}_t) = \mathbf{A}_{e,k} \text{mat}_k(\mathcal{F}_{e,t}) \mathbf{A}'_{e,-k} + \text{mat}_k(\boldsymbol{\Sigma}_{\epsilon}) \circ \text{mat}_k(\boldsymbol{\epsilon}_t),$$

where $\mathbf{A}_{e,-k} := \mathbf{A}_{e,K} \otimes \dots \otimes \mathbf{A}_{e,k+1} \otimes \mathbf{A}_{e,k-1} \otimes \dots \otimes \mathbf{A}_{e,1}$. Each mode- k noise fibre $\mathbf{e}_{t,k,l}$ for $l \in [d_k]$ can then be decomposed as

$$\mathbf{e}_{t,k,l} := \mathbf{A}_{e,k} \text{mat}_k(\mathcal{F}_{e,t}) \mathbf{A}_{e,-k,l} + \boldsymbol{\Sigma}_{\epsilon,k,l}^{1/2} \boldsymbol{\epsilon}_{t,k,l}, \quad (\text{S.3})$$

where $\boldsymbol{\Sigma}_{\epsilon,k,l} = \text{diag}((\text{mat}_k(\boldsymbol{\Sigma}_{\epsilon}))_{\cdot l} (\text{mat}_k(\boldsymbol{\Sigma}_{\epsilon}))'_{\cdot l})$, and $\boldsymbol{\epsilon}_{t,k,l}$ contains independent elements each with mean 0 and variance 1.

Given $h \in [d_k]$, $i \in [d_k]$, from (S.3) and Assumption (E1) and (E2), we have

$$\sum_{l \neq h} \sum_{j=1}^{d_k} \left| \mathbb{E}[\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jl}] \right| \leq \|\mathbf{A}_{e,-k,h}\| \|\mathbf{A}_{e,-k,l}\| \|\mathbf{A}_{e,k}\|_1 \|\mathbf{A}_{e,k}\|_{\infty} = O(1).$$

Moreover,

$$\begin{aligned} \sum_{j=1}^{d_k} \left| \mathbb{E}[\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jh}] \right| &\leq \|\text{cov}(\mathbf{e}_{t,k,h}, \mathbf{e}_{t,k,h})\|_1 \\ &\leq \|\mathbf{A}_{e,-k,h}\|^2 \|\mathbf{A}_{e,k}\|_1 \|\mathbf{A}_{e,k}\|_{\infty} + \|\boldsymbol{\Sigma}_{\epsilon,k,h}\|_1 = O(1), \end{aligned}$$

where the last equality is from Assumption (E1).

To finish the proof of the first result in the Proposition, fix indices k, t, i, j, h . From Assumption (E2) and (S.3), we have

$$[\text{mat}_k(\mathcal{E}_t)]_{il} = \sum_{q \geq 0} a_{\epsilon, q} \mathbf{A}'_{e,k,i} \cdot \text{mat}_k(\mathcal{X}_{\epsilon, t-q}) \mathbf{A}_{e,-k,l} + [\text{mat}_k(\boldsymbol{\Sigma}_{\epsilon})]_{il} \sum_{q \geq 0} a_{\epsilon, q} \text{mat}_k(\mathcal{X}_{\epsilon, t-q})_{il}. \quad (\text{S.4})$$

Hence when $l \neq h$, from the independence between $\{\mathcal{X}_{\epsilon, t}\}$ and $\{\mathcal{X}_{\epsilon, s}\}$ and the independence of the elements within $\{\mathcal{X}_{\epsilon, t}\}$ in Assumption (E2), we have for any $s \in \psi_{k, ij, l}$,

$$\begin{aligned} &\text{cov}\left(\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{il} \text{mat}_k(\mathcal{E}_s)_{jl}\right) \\ &= \text{cov}\left(\left(\sum_{q \geq 0} a_{\epsilon, q} \mathbf{A}'_{e,k,i} \cdot \text{mat}_k(\mathcal{X}_{\epsilon, t-q}) \mathbf{A}_{e,-k,h}\right) \left(\sum_{q \geq 0} a_{\epsilon, q} \mathbf{A}'_{e,k,j} \cdot \text{mat}_k(\mathcal{X}_{\epsilon, t-q}) \mathbf{A}_{e,-k,h}\right), \right. \\ &\quad \left. \left(\sum_{q \geq 0} a_{\epsilon, q} \mathbf{A}'_{e,k,i} \cdot \text{mat}_k(\mathcal{X}_{\epsilon, s-q}) \mathbf{A}_{e,-k,l}\right) \left(\sum_{q \geq 0} a_{\epsilon, q} \mathbf{A}'_{e,k,j} \cdot \text{mat}_k(\mathcal{X}_{\epsilon, s-q}) \mathbf{A}_{e,-k,l}\right)\right). \end{aligned} \quad (\text{S.5})$$

By (E2) again, all mixed covariance terms are zero except for $\text{cov}(\text{mat}_k(\mathcal{X}_{e,t-q})_{nm}^2, \text{mat}_k(\mathcal{X}_{e,t-q})_{nm}^2)$ for all $q \geq 0, n \in [r_{e,k}], m \in [r_{e,-k}]$, with coefficient $a_{e,q}^2 a_{e,q+|t-s|}^2 A_{e,k,in}^2 A_{e,k,jn}^2 A_{e,-k,hm}^2 A_{e,-k,lm}^2$. Thus we have

$$\begin{aligned} & \sum_{l=1, l \neq h}^{d_k} \left| \text{cov}(\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{il} \text{mat}_k(\mathcal{E}_s)_{jl}) \right| \\ &= \sum_{l=1, l \neq h}^{d_k} \left| \sum_{n=1}^{r_{e,k}} \sum_{m=1}^{r_{e,-k}} \sum_{q \geq 0} a_{e,q}^2 a_{e,q+|t-s|}^2 A_{e,k,in}^2 A_{e,k,jn}^2 A_{e,-k,hm}^2 A_{e,-k,lm}^2 \cdot \text{cov}(\text{mat}_k(\mathcal{X}_{e,t-q})_{nm}^2, \text{mat}_k(\mathcal{X}_{e,t-q})_{nm}^2) \right| \\ &= O(1) \cdot \sum_{q \geq 0} a_{e,q}^2 a_{e,q+|t-s|}^2 \left(\sum_{n=1}^{r_{e,k}} A_{e,k,in}^2 A_{e,k,jn}^2 \right) \left(\sum_{m=1}^{r_{e,-k}} A_{e,-k,hm}^2 \sum_{l=1, l \neq h}^{d_k} A_{e,-k,lm}^2 \right) = \sum_{q \geq 0} O(a_{e,q}^2 a_{e,q+|t-s|}^2), \end{aligned}$$

where we use Assumption (E2) in the second last equality, and (E1) in the last. Consequently,

$$\sum_{l=1, l \neq h}^{d_k} \sum_{s \in \psi_{k,ij,l}} \left| \text{cov}(\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{il} \text{mat}_k(\mathcal{E}_s)_{jl}) \right| = \sum_{q \geq 0} \sum_{s=1}^T O(a_{e,q}^2 a_{e,q+|t-s|}^2) = O(1),$$

where the last equality uses Assumption (E2). Now consider lastly $l = h$. All arguments starting from (S.4) follow exactly, except the following term is added in (S.5):

$$\begin{aligned} & \sum_{q \geq 0} a_{e,q}^2 a_{e,q+|t-s|}^2 \Sigma_{\epsilon,k,h,ii} \Sigma_{\epsilon,k,h,jj} \cdot \text{cov}(\text{mat}_k(\mathcal{X}_{e,t-q})_{ih} \text{mat}_k(\mathcal{X}_{e,t-q})_{jh}, \text{mat}_k(\mathcal{X}_{e,s-q})_{ih} \text{mat}_k(\mathcal{X}_{e,s-q})_{jh}) \\ &= O(1) \cdot \sum_{q \geq 0} a_{e,q}^2 a_{e,q+|t-s|}^2 = O(1). \end{aligned}$$

where we used again Assumptions (E2) in the last line. Finally,

$$\sum_{s \in \psi_{k,ij,h}} \left| \text{cov}(\text{mat}_k(\mathcal{E}_t)_{ih} \text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{ih} \text{mat}_k(\mathcal{E}_s)_{jh}) \right| = O(1).$$

This completes the proof of result 1 in the Proposition.

To prove the second result, fix $k \in [K], i, j \in [d_k]$ and deterministic vectors $\mathbf{u} \in \mathbb{R}^{r_k}$ and $\mathbf{v} \in \mathbb{R}^{r_{-k}}$ with $\|\mathbf{u}\|, \|\mathbf{v}\| = O(1)$. Note that $\mathbb{E}[\text{mat}_k(\mathcal{F}_t) \mathbf{v} \mathbf{v}' \text{mat}_k(\mathcal{F}_s)'] = \mathbf{v}' \mathbf{v} (r_{-k} \sum_{q \geq 0} a_{f,q} a_{f,q+|t-s|}) \mathbf{I}_{r_k}$, as the series $\{\mathcal{X}_{f,t}\}$ has i.i.d. elements from Assumption (F1). Similarly, from (S.3) and Assumption (E1) and (E2),

$$\begin{aligned} & \text{cov}(\text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{jl}) \\ &= \mathbb{E}[\mathbf{A}'_{e,k,j} \cdot \text{mat}_k(\mathcal{F}_{e,t}) \mathbf{A}_{e,-k,h} \cdot \mathbf{A}'_{e,-k,l} \cdot \text{mat}_k(\mathcal{F}_{e,s})' \mathbf{A}_{e,k,j}] + \mathbb{E}[\epsilon'_{t,k,h} (\Sigma_{\epsilon,k,h,j} \cdot \Sigma'_{\epsilon,k,l,j})^{1/2} \epsilon_{s,k,l}] \\ &= \mathbf{A}'_{e,-k,l} \cdot \mathbf{A}_{e,-k,h} \cdot \|\mathbf{A}_{e,k,j}\|^2 \cdot \sum_{q \geq 0} a_{e,q} a_{e,q+|t-s|} + \mathbb{1}_{\{h=l\}} \cdot \Sigma_{\epsilon,k,h,jj} \sum_{q \geq 0} a_{e,q} a_{e,q+|t-s|}. \end{aligned}$$

Hence if we fix $h \in [d_k]$, $t \in \psi_{k,ij,h}$, then together with Assumption (E2), we have

$$\begin{aligned}
& \sum_{l=1}^{d_k} \sum_{s \in \psi_{k,ij,l}} \frac{1}{|\psi_{k,ij,l}|} \cdot \mathbb{E} \left[\text{mat}_k(\mathcal{E}_t)_{jh} \mathbf{u}' \text{mat}_k(\mathcal{F}_t) \mathbf{v} \cdot \text{mat}_k(\mathcal{E}_s)_{jl} \mathbf{v}' \text{mat}_k(\mathcal{F}_s)' \mathbf{u} \right] \\
&= \sum_{l=1}^{d_k} \sum_{s \in \psi_{k,ij,l}} \frac{1}{|\psi_{k,ij,l}|} \cdot \text{cov}(\text{mat}_k(\mathcal{E}_t)_{jh}, \text{mat}_k(\mathcal{E}_s)_{jl}) \cdot \mathbb{E} \left[\mathbf{u}' \text{mat}_k(\mathcal{F}_t) \mathbf{v} \mathbf{v}' \text{mat}_k(\mathcal{F}_s)' \mathbf{u} \right] \\
&= \sum_{l=1}^{d_k} \frac{1}{|\psi_{k,ij,l}|} \left[O(\mathbf{A}'_{e,-k,l} \mathbf{A}_{e,-k,h} \cdot \|\mathbf{A}_{e,k,j}\|^2) \cdot \sum_{q \geq 0} \sum_{p \geq 0} \sum_{s \in \psi_{k,ij,l}} a_{e,q} a_{e,q+|t-s|} a_{f,p} a_{f,p+|t-s|} \right. \\
&\quad \left. + O(\mathbb{1}_{\{h=l\}} \cdot \Sigma_{\epsilon,k,h,jj}) \cdot \sum_{q \geq 0} \sum_{p \geq 0} \sum_{s \in \psi_{k,ij,l}} a_{\epsilon,q} a_{\epsilon,q+|t-s|} a_{f,p} a_{f,p+|t-s|} \right] \\
&= \sum_{l=1}^{d_k} \frac{1}{|\psi_{k,ij,l}|} \cdot O(\mathbf{A}'_{e,-k,l} \mathbf{A}_{e,-k,h} \cdot \|\mathbf{A}_{e,k,j}\|^2 + \mathbb{1}_{\{h=l\}} \cdot \Sigma_{\epsilon,k,h,jj}) = O\left(\frac{1}{T}\right),
\end{aligned} \tag{S.6}$$

where for the second last equality, we argue for the first term in the second last line only, as the second term could be shown similarly:

$$\begin{aligned}
& \sum_{q \geq 0} \sum_{p \geq 0} \sum_{s \in \psi_{k,ij,h}} a_{e,q} a_{e,q+|t-s|} a_{f,p} a_{f,p+|t-s|} = \sum_{q \geq 0} \sum_{p \geq 0} a_{e,q} a_{f,p} \sum_{s \in \psi_{k,ij,h}} a_{e,q+|t-s|} a_{f,p+|t-s|} \\
&\leq \sum_{q \geq 0} \sum_{p \geq 0} |a_{e,q}| \cdot |a_{f,p}| \cdot \left(\sum_{s \in \psi_{k,ij,h}} a_{e,q+|t-s|}^2 \right)^{1/2} \left(\sum_{s \in \psi_{k,ij,h}} a_{f,p+|t-s|}^2 \right)^{1/2} \leq \sum_{q \geq 0} \sum_{p \geq 0} |a_{e,q}| \cdot |a_{f,p}| \leq c^2,
\end{aligned}$$

where the constant c is from Assumptions (F1) and (E2). Finally,

$$\begin{aligned}
& \mathbb{E} \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{k,ij,h}} \frac{1}{d_k \cdot |\psi_{k,ij,h}|} \text{mat}_k(\mathcal{E}_t)_{jh} \mathbf{u}' \text{mat}_k(\mathcal{F}_t) \mathbf{v} \right)^2 \\
&= \frac{1}{d_k^2} \sum_{h=1}^{d_k} \sum_{l=1}^{d_k} \sum_{t \in \psi_{k,ij,h}} \sum_{s \in \psi_{k,ij,l}} \frac{1}{|\psi_{k,ij,h}| \cdot |\psi_{k,ij,l}|} \mathbb{E} \left[\text{mat}_k(\mathcal{E}_t)_{jh} \mathbf{u}' \text{mat}_k(\mathcal{F}_t) \mathbf{v} \cdot \text{mat}_k(\mathcal{E}_s)_{jl} \mathbf{v}' \text{mat}_k(\mathcal{F}_s)' \mathbf{u} \right] \\
&= \frac{1}{d_k^2 T} \sum_{h=1}^{d_k} \sum_{t \in \psi_{k,ij,h}} O\left(\frac{1}{T}\right) = O\left(\frac{1}{d_k T}\right),
\end{aligned}$$

which then implies result 2 of the Proposition.

Finally, we prove result 3 of the Proposition. From Assumption (F1), we have $\mathbb{E}[\mathcal{F}_t] = 0$. Next, for any $t \in [T]$, it is direct from Assumption (F1) that all elements in $\mathcal{F}_t$ are independent. Moreover,

$$\begin{aligned}
\mathbb{E}[\text{mat}_k(\mathcal{F}_t) \text{mat}_k(\mathcal{F}_t)'] &= \mathbb{E} \left[\left(\sum_{q \geq 0} a_{f,q} \text{mat}_k(\mathcal{X}_{f,t-q}) \right) \left(\sum_{q \geq 0} a_{f,q} \text{mat}_k(\mathcal{X}_{f,t-q})' \right) \right] \\
&= \sum_{q \geq 0} a_{f,q}^2 \mathbb{E}[\text{mat}_k(\mathcal{X}_{f,t-q}) \text{mat}_k(\mathcal{X}_{f,t-q})'] = \left(\sum_{q \geq 0} a_{f,q}^2 \right) \cdot r_{-k} \mathbf{I}_{r_k} = r_{-k} \mathbf{I}_{r_k},
\end{aligned}$$

where we use Assumption (F1) in the last line. To complete the proof, WLOG consider the variance of the

j -th diagonal element of $\text{mat}_k(\mathcal{F}_t)\text{mat}_k(\mathcal{F}_t)'$. From Assumption (F1), we have

$$\begin{aligned}
\text{Var}\left(\frac{1}{T}\sum_{t=1}^T[\text{mat}_k(\mathcal{F}_t)]'_j.[\text{mat}_k(\mathcal{F}_t)]_j\right) &= \frac{1}{T^2}\text{Var}\left(\sum_{t=1}^T\left[\sum_{q\geq 0}a_{f,q}[\text{mat}_k(\mathcal{X}_{f,t-q})]'_j\right]\left[\sum_{q\geq 0}a_{f,q}[\text{mat}_k(\mathcal{X}_{f,t-q})]_j\right]\right) \\
&= \frac{1}{T^2}\sum_{t=1}^T\sum_{s=1}^T\text{cov}\left(\left[\sum_{q\geq 0}a_{f,q}[\text{mat}_k(\mathcal{X}_{f,t-q})]'_j\right]\left[\sum_{q\geq 0}a_{f,q}[\text{mat}_k(\mathcal{X}_{f,t-q})]_j\right],\right. \\
&\quad \left.\left[\sum_{q\geq 0}a_{f,q}[\text{mat}_k(\mathcal{X}_{f,s-q})]'_j\right]\left[\sum_{q\geq 0}a_{f,q}[\text{mat}_k(\mathcal{X}_{f,s-q})]_j\right]\right) \\
&= \frac{1}{T^2}\sum_{t=1}^T\sum_{q\geq 0}a_{f,q}^4\text{Var}([\text{mat}_k(\mathcal{X}_{f,t-q})]'_j.[\text{mat}_k(\mathcal{X}_{f,t-q})]_j) \\
&+ \frac{1}{T^2}\sum_{t=1}^T\sum_{q\geq 0}\sum_{p\neq q}a_{f,q}^2a_{f,p}^2\text{Var}([\text{mat}_k(\mathcal{X}_{f,t-q})]'_j.[\text{mat}_k(\mathcal{X}_{f,t-p})]_j) \\
&= \frac{r-k}{T^2}\sum_{t=1}^T\sum_{q\geq 0}a_{f,q}^4\text{Var}([\text{mat}_k(\mathcal{X}_{f,t-q})]_1^2) + \frac{r-k}{T^2}\sum_{t=1}^T\sum_{q\geq 0}\sum_{p\neq q}a_{f,q}^2a_{f,p}^2 \\
&= O(1) \cdot \frac{1}{T^2}\sum_{t=1}^T\sum_{q\geq 0}a_{f,q}^4 + O(1) \cdot \frac{1}{T^2}\sum_{t=1}^T\sum_{q\geq 0}\sum_{p\neq q}a_{f,q}^2a_{f,p}^2 = O(1) \cdot \frac{1}{T^2}\sum_{t=1}^T(\sum_{q\geq 0}a_{f,q}^2)^2 = O\left(\frac{1}{T}\right) = o(1),
\end{aligned}$$

where the third equality uses the independence in Assumption (E2). This completes the proof of result 3, and hence the Proposition. $\square$

To prove Theorem 1, we are going to present some lemmas and prove them first. From (S.1),

$$\hat{\mathbf{Q}}_{j\cdot} = \hat{\mathbf{D}}^{-1}\sum_{i=1}^{d_k}\hat{\mathbf{Q}}_{i\cdot}\hat{S}_{ij} = \hat{\mathbf{D}}^{-1}\sum_{i=1}^{d_k}\hat{\mathbf{Q}}_{i\cdot}\sum_{h=1}^{d_k}\frac{1}{|\psi_{ij,h}|}\sum_{t\in\psi_{ij,h}}Y_{t,ih}Y_{t,jh}. \quad (\text{S.7})$$

With the notations in (S.2), (3.1) can be written as $\mathbf{Y}_t = \mathbf{Q}\mathbf{F}_{Z,t}\mathbf{A}' + \mathbf{E}_t$, and hence for $i, j \in [d_k], h \in [d_k]$,

$$\begin{aligned}
Y_{t,ih} &= \left(\sum_{n=1}^{r_k}\sum_{m=1}^{r-k}Q_{in}\Lambda_{hm}F_{Z,t,nm}\right) + E_{t,ih} = \mathbf{Q}'_{i\cdot}\left(\sum_{m=1}^{r-k}\Lambda_{hm}\mathbf{F}_{Z,t,\cdot m}\right) + E_{t,ih} \\
&= \left(\sum_{m=1}^{r-k}\Lambda_{hm}\mathbf{F}_{Z,t,\cdot m}\right)'\mathbf{Q}_{i\cdot} + E_{t,ih}.
\end{aligned}$$

Hence the product $Y_{t,ih}Y_{t,jh}$ in (S.7) can be written as

$$\begin{aligned}
Y_{t,ih}Y_{t,jh} &= \left(\sum_{m=1}^{r-k}\Lambda_{hm}\mathbf{F}_{Z,t,\cdot m}\right)'\mathbf{Q}_{i\cdot}\left(\sum_{m=1}^{r-k}\Lambda_{hm}\mathbf{F}_{Z,t,\cdot m}\right)'\mathbf{Q}_{j\cdot} + E_{t,ih}E_{t,jh} \\
&\quad + E_{t,jh}\left(\sum_{m=1}^{r-k}\Lambda_{hm}\mathbf{F}_{Z,t,\cdot m}\right)'\mathbf{Q}_{i\cdot} + E_{t,ih}\left(\sum_{m=1}^{r-k}\Lambda_{hm}\mathbf{F}_{Z,t,\cdot m}\right)'\mathbf{Q}_{j\cdot}.
\end{aligned} \quad (\text{S.8})$$

We then have, from (S.7) and (S.8) that

$$\begin{aligned}
\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot} &= \widehat{\mathbf{D}}^{-1} \left(\sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right. \\
&\quad + \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{i\cdot} \\
&\quad \left. + \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{j\cdot} \right) \\
&=: \widehat{\mathbf{D}}^{-1} (\mathcal{I}_j + \mathcal{II}_j + \mathcal{III}_j),
\end{aligned} \tag{S.9}$$

where

$$\begin{aligned}
\mathcal{I}_j &:= \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh}, \\
\mathcal{II}_j &:= \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{i\cdot}, \\
\mathcal{III}_j &:= \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{j\cdot}.
\end{aligned}$$

The following lemma bounds the terms $\mathcal{I}_j$, $\mathcal{II}_j$, and $\mathcal{III}_j$.

Lemma 1 *Under Assumptions (O1), (F1), (L1), (E1) and (E2), we have*

$$\frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathcal{I}_j\|_F^2 = O_P\left(\frac{d}{T} + d_{-k}^2\right), \tag{S.10}$$

$$\frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathcal{II}_j\|_F^2 = O_P\left(\frac{d_{-k} d_k^{\alpha_{k,1}}}{T}\right) = \frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathcal{III}_j\|_F^2. \tag{S.11}$$

Proof of Lemma 1. To prove (S.10), we decompose

$$\begin{aligned}
\mathcal{I}_j &= \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh}}{|\psi_{ij,h}|} \\
&= \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \left(\frac{\sum_{t \in \psi_{ij,h}} (E_{t,ih} E_{t,jh} - \mathbb{E}[E_{t,ih} E_{t,jh}])}{|\psi_{ij,h}|} + \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih} E_{t,jh}]}{|\psi_{ij,h}|} \right) \\
&=: \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \xi_{ij} + \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \eta_{ij},
\end{aligned} \tag{S.12}$$

where

$$\xi_{ij} := \sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} (E_{t,ih} E_{t,jh} - \mathbb{E}[E_{t,ih} E_{t,jh}])}{|\psi_{ij,h}|}, \quad \eta_{ij} := \sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih} E_{t,jh}]}{|\psi_{ij,h}|}.$$

We want to show the following:

$$\sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \xi_{ij} \right\|_F^2 = O_P\left(\frac{dd_k}{T}\right), \tag{S.13}$$

$$\sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \eta_{ij} \right\|_F^2 = O_P(dd_{-k}). \tag{S.14}$$

To show (S.13), first note that $\mathbb{E}\xi_{ij} = 0$, and also by Assumption (O1),

$$\begin{aligned}
\mathbb{E}|\xi_{ij}|^2 &= \text{var}(\xi_{ij}) \leq \frac{1}{\psi_0^2 T^2} \text{var} \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} (E_{t,ih} E_{t,jh} - \mathbb{E}[E_{t,ih} E_{t,jh}]) \right) \\
&\leq \frac{1}{\psi_0^2 T^2} \sum_{h=1}^{d_k} \sum_{l=1}^{d_k} \sum_{t \in \psi_{ij,h}} \sum_{s \in \psi_{ij,l}} \left| \text{cov} \left((E_{t,ih} E_{t,jh} - \mathbb{E}[E_{t,ih} E_{t,jh}]), (E_{s,il} E_{s,jl} - \mathbb{E}[E_{s,il} E_{s,jl}]) \right) \right| \\
&= \frac{1}{\psi_0^2 T^2} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \sum_{l=1}^{d_k} \sum_{s \in \psi_{ij,l}} \left| \text{cov} \left(E_{t,ih} E_{t,jh}, E_{s,il} E_{s,jl} \right) \right| \leq \frac{cd_k}{\psi_0^2 T}, \tag{S.15}
\end{aligned}$$

where the last inequality and the constant c are from result 1 of Proposition 1 (hereafter Proposition 1.1, etc). Then by the Cauchy–Schwarz inequality,

$$\sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i \cdot} \xi_{ij} \right\|_F^2 \leq \sum_{j=1}^{d_k} \left(\sum_{i=1}^{d_k} \|\hat{\mathbf{Q}}_{i \cdot}\|_F^2 \right) \left(\sum_{i=1}^{d_k} \xi_{ij}^2 \right) = O_P \left(\frac{dd_k}{T} \right),$$

which is (S.13). To show (S.14), note that if we define

$$\rho_{ij,h} := \frac{\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih} E_{t,jh}]}{\left(\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih}^2] \right)^{1/2} \left(\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,jh}^2] \right)^{1/2}},$$

then $|\rho_{ij,h}| < 1$ and hence $\rho_{ij,h}^2 \leq |\rho_{ij,h}|$. It is then easy to prove also that

$$\rho_{ij} := \frac{\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih} E_{t,jh}]}{\left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih}^2] \right)^{1/2} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,jh}^2] \right)^{1/2}},$$

also satisfy $|\rho_{ij}| \leq 1$ and $\rho_{ij}^2 \leq |\rho_{ij}|$. By Proposition 1.1,

$$\left| \sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih}^2]}{|\psi_{ij,h}|} \right| = O_P(d_k),$$

and hence

$$\begin{aligned}
\eta_{ij}^2 &= \left(\sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih} E_{t,jh}]}{|\psi_{ij,h}|} \right)^2 = \rho_{ij}^2 \left(\sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih}^2]}{|\psi_{ij,h}|} \right) \left(\sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,jh}^2]}{|\psi_{ij,h}|} \right) \\
&= |\rho_{ij}| \left(\sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih}^2]}{|\psi_{ij,h}|} \right)^{1/2} \left(\sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,jh}^2]}{|\psi_{ij,h}|} \right)^{1/2} \cdot O_P(d_k) \\
&= \left| \sum_{h=1}^{d_k} \frac{\sum_{t \in \psi_{ij,h}} \mathbb{E}[E_{t,ih} E_{t,jh}]}{|\psi_{ij,h}|} \right| \cdot O_P(d_k) = |\eta_{ij}| O_P(d_k).
\end{aligned}$$

Using the above, we then have

$$\begin{aligned}
\sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i \cdot} \eta_{ij} \right\|_F^2 &\leq \sum_{j=1}^{d_k} \left(\sum_{i=1}^{d_k} \|\hat{\mathbf{Q}}_{i \cdot}\|_F^2 \right) \left(\sum_{i=1}^{d_k} \eta_{ij}^2 \right) \leq r_k \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \eta_{ij}^2 = O_P(d_k) \cdot \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} |\eta_{ij}| \\
&= O_P(d_k) \cdot \frac{1}{|\psi_0 T|} \sum_{t \in \psi_{ij,h}} \sum_{h=1}^{d_k} \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \left| \mathbb{E}[E_{t,ih} E_{t,jh}] \right| = O_P(dd_k), \tag{S.16}
\end{aligned}$$

where the second last equality used Assumption (O1), and the last equality used Proposition 1.1. This proves (S.14). Using (S.13) and (S.14), from (S.12) we have

$$\begin{aligned} \frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathcal{I}_j\|_F^2 &= \frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \xi_{ij,h} + \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \eta_{ij,h} \right\|_F^2 \\ &\leq \frac{2}{d_k} \sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i\cdot} \xi_{ij} \right\|_F^2 + \frac{2}{d_k} \sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i\cdot} \eta_{ij} \right\|_F^2 = O_P\left(\frac{d}{T} + d_k^2\right). \end{aligned}$$

This completes the proof of (S.10). To prove (S.11), consider

$$\begin{aligned} \frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathcal{II}_j\|_F^2 &= \frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{i\cdot} \right\|_F^2 \\ &\leq \frac{1}{d_k} \sum_{j=1}^{d_k} \left(\sum_{i=1}^{d_k} \|\hat{\mathbf{Q}}_{i\cdot}\|_F^2 \right) \cdot \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{i\cdot} \right)^2 \\ &= \frac{r_k}{d_k} \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r-k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot m} \right)' \mathbf{Q}_{i\cdot} \right)^2 \\ &= \frac{r_k d_k^2}{d_k} \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \frac{1}{d_k \cdot |\psi_{ij,h}|} E_{t,jh} [\otimes_{l \in [K] \setminus \{k\}} \mathbf{A}_l]'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \right)^2 \\ &= \frac{r_k d_k^2}{d_k} \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \|\mathbf{u}_i\|_F^2 \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \frac{1}{d_k \cdot |\psi_{ij,h}|} E_{t,jh} \mathbf{v}'_h \mathbf{F}'_t \frac{1}{\|\mathbf{u}_i\|_F} \mathbf{u}_i \right)^2, \end{aligned} \tag{S.17}$$

where $\mathbf{A} = \mathbf{A}_k$ and $\mathbf{F}_t = \text{mat}_k(\mathcal{F}_t)$ above, and we define $\mathbf{v}_h := [\otimes_{l \in [K] \setminus \{k\}} \mathbf{A}_l]_h$, $\mathbf{u}_i := \mathbf{A}_{i\cdot}$. By Proposition 1.2, the last bracket in the last line of (S.17) is $O_P(d_k^{-1} T^{-1})$, and hence

$$\frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathcal{II}_j\|_F^2 = O_P\left(\frac{d_k}{d_k T}\right) \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \|\mathbf{u}_i\|_F^2 = O_P\left(\frac{d_k}{d_k T}\right) \sum_{j=1}^{d_k} \|\mathbf{A}\|_F^2 = O_P\left(\frac{d_k d_k^{\alpha_{k,1}}}{T}\right),$$

where the last equality follows since for any $l \in [K]$, $\|\mathbf{A}_l\|_F^2 = O_P(\text{tr}(\mathbf{Z}_l)) = O_P(d_l^{\alpha_{l,1}})$ by Assumption (L1). The bound corresponding to $\mathcal{III}_j$ can be proved similarly (omitted), and hence (S.11) is established. This concludes the proof of Lemma 1. $\square$

Lemma 2 Under Assumptions (O1), (M1), (F1), (L1), (E1), (E2) and (R1), with $\mathbf{H}_j$ and $\hat{\mathbf{D}}$ from (S.2), we have

$$\|\hat{\mathbf{D}}^{-1}\|_F = O_P\left(d_k^{\alpha_{k,1} - \alpha_{k,r_k}} \prod_{j=1}^K d_j^{-\alpha_{j,1}}\right), \tag{S.18}$$

$$\frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \hat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot} \right\|_F^2 = O_P\left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) - 1} \left(\frac{1}{T d_k} + \frac{1}{d_k}\right) \prod_{j=1}^K d_j^{2(1 - \alpha_{j,1})}\right). \tag{S.19}$$

Proof of Lemma 2. First, we bound the term $\|\hat{\mathbf{D}}^{-1}\|_F^2$ by finding the lower bound of $\lambda_{r_k}(\hat{\mathbf{D}})$. To do this, define $\omega_k := d_k^{\alpha_{k,r_k} - \alpha_{k,1}} \prod_{j=1}^K d_j^{\alpha_{j,1}}$, and consider the decomposition

$$\hat{\mathbf{S}} = \mathbf{R}^* + (\tilde{\mathbf{R}} - \mathbf{R}^*) + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3, \tag{S.20}$$

where for a unit vector γ ,

$$\begin{aligned}
R(\gamma) &:= \frac{1}{\omega_k} \gamma' \widehat{\mathbf{S}} \gamma = \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \widehat{S}_{ij} = \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} Y_{t,ih} Y_{t,jh} \\
&=: R^*(\gamma) + (\widetilde{R}(\gamma) - R^*(\gamma)) + R_1 + R_2 + R_3, \quad \text{with} \\
\widetilde{R}(\gamma) &:= \frac{1}{\omega_k} \gamma' \widetilde{\mathbf{R}} \gamma := \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_i \cdot \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_j, \\
R^*(\gamma) &:= \frac{1}{\omega_k} \gamma' \mathbf{R}^* \gamma := \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \frac{1}{T} \sum_{t=1}^T \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_i \cdot \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_j, \\
R_1 &:= \frac{1}{\omega_k} \gamma' \mathbf{R}_1 \gamma := \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh}, \\
R_2 &:= \frac{1}{\omega_k} \gamma' \mathbf{R}_2 \gamma := \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_i, \\
R_3 &:= \frac{1}{\omega_k} \gamma' \mathbf{R}_3 \gamma := \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_j, \tag{S.21}
\end{aligned}$$

and we used (S.8) for the expansion above. Then we have the decomposition

$$R(\gamma) - R^*(\gamma) = R(\gamma) - \widetilde{R}(\gamma) + \widetilde{R}(\gamma) - R^*(\gamma). \tag{S.22}$$

Similar to the treatment of the term $\mathcal{I}_j$ in the proof of Lemma 1, since $\|\gamma\| = 1$,

$$\begin{aligned}
|R_1| &\leq \frac{1}{\omega_k} \left| \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \xi_{ij} \right| + \frac{1}{\omega_k} \left| \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \eta_{ij} \right| \leq \frac{1}{\omega_k} \left(\sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \xi_{ij}^2 \right)^{1/2} + \frac{1}{\omega_k} \left(\sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \eta_{ij}^2 \right)^{1/2} \\
&= O_P \left(d_k^{\alpha_k, 1 - \alpha_k, r_k} \left[\frac{1}{T^{1/2} d_k^{1/2}} + \frac{1}{d_k^{1/2}} \right] \prod_{j=1}^K d_j^{1 - \alpha_{j,1}} \right) = O_P(d[(T d_k)^{-1/2} + d_k^{-1/2}]/\omega_k), \tag{S.23}
\end{aligned}$$

where the second last equality is from (S.15) and part of (S.16). Together with Assumption (R1), (S.23) implies that as $T, d_k, d_{-k} \rightarrow \infty$, $R_1 \xrightarrow{P} 0$.

From (S.17) and the arguments for $\mathcal{I}_j$ immediately afterwards, we see that

$$\begin{aligned}
|R_2| &\leq \frac{1}{\omega_k} \left\{ \sum_{j=1}^{d_k} \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} \left(\sum_{m=1}^{r_k} \Lambda_{hm} \mathbf{F}_{Z,t,\cdot,m} \right)' \mathbf{Q}_i \right)^2 \right\}^{1/2} \\
&= O_P(\omega_k^{-1}) \cdot O_P \left(T^{-1/2} d^{1/2} \prod_{j=1}^K d_j^{\alpha_{j,1}/2} \right) = O_P((dg_s)^{1/2} T^{-1/2} / \omega_k) = o_P(1), \tag{S.24}
\end{aligned}$$

where the last equality is from Assumption (R1). The term R_3 can be proved to have the same rate with same lines of proof as for R_2 . Hence we have

$$\sup_{\|\gamma\|=1} |R(\gamma) - \widetilde{R}(\gamma)| = \sup_{\|\gamma\|=1} |R_1 + R_2 + R_3| \xrightarrow{P} 0. \tag{S.25}$$

Similar to the proof of (R6) in Lemma 4 in Xiong and Pelger (2023), using the definition $\mathbf{v}_h :=$

$[\otimes_{l \in [K] \setminus \{k\}} \mathbf{A}_l]_h$. as before,

$$\begin{aligned}
& \tilde{R}(\gamma) - R^*(\gamma) \\
&= \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \left(\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{Q}'_i \mathbf{F}_{Z,t} \mathbf{\Lambda}_h \mathbf{\Lambda}'_h \mathbf{F}'_{Z,t} \mathbf{Q}_j - \frac{1}{T} \sum_{t=1}^T \mathbf{Q}'_i \mathbf{F}_{Z,t} \mathbf{\Lambda}_h \mathbf{\Lambda}'_h \mathbf{F}'_{Z,t} \mathbf{Q}_j \right) \\
&= \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \mathbf{A}'_i \left(\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{F}_t \mathbf{v}_h \mathbf{v}'_h \mathbf{F}'_t - \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{v}_h \mathbf{v}'_h \mathbf{F}'_t \right) \mathbf{A}_j. \\
&=: \frac{1}{\omega_k} \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \gamma_i \gamma_j \sum_{h=1}^{d_k} \mathbf{A}'_i \Delta_{F,k,ij,h} \mathbf{A}_j, \quad \text{where} \tag{S.26} \\
& \Delta_{F,k,ij,h} := \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{F}_t \mathbf{v}_h \mathbf{v}'_h \mathbf{F}'_t - \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{v}_h \mathbf{v}'_h \mathbf{F}'_t.
\end{aligned}$$

By the Cauchy–Schwarz inequality, we then have

$$|\tilde{R}(\gamma) - R^*(\gamma)| \leq \frac{1}{\omega_k} \left(\sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \left[\mathbf{A}'_i \left(\sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \right) \mathbf{A}_j \right]^2 \right)^{1/2}. \tag{S.27}$$

With Assumption (M1), using the standard rate of convergence in the weak law of large number for α -mixing sequence and the fact that the elements in $\mathcal{F}_t$ are independent from Assumption (F1), since $\Delta_{F,k,ij,h}$ has fixed dimension, we have for each $k \in [K]$, $i, j \in [d_k]$ and $h \in [d_k]$,

$$\|\Delta_{F,k,ij,h}\|_F \leq \left\| \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{F}_t \mathbf{v}_h \mathbf{v}'_h \mathbf{F}'_t - \mathbf{v}'_h \mathbf{v}_h \Sigma_k \right\|_F + \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{v}_h \mathbf{v}'_h \mathbf{F}'_t - \mathbf{v}'_h \mathbf{v}_h \Sigma_k \right\|_F = O_P\left(\frac{\|\mathbf{v}_h\|^2}{T^{1/2}}\right). \tag{S.28}$$

From Assumption (L1), we then have

$$\begin{aligned}
\sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \left[\mathbf{A}'_i \left(\sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \right) \mathbf{A}_j \right]^2 &\leq \sum_{i=1}^{d_k} \sum_{j=1}^{d_k} \|\mathbf{A}_i\|_F^2 \|\mathbf{A}_j\|_F^2 \left(\sum_{h=1}^{d_k} \|\Delta_{F,k,ij,h}\|_F \right)^2 \\
&\leq \|\mathbf{A}_k\|_F^4 \cdot O_P\left(\frac{1}{T} \left(\sum_{h=1}^{d_k} \|\mathbf{v}_h\|^2 \right)^2\right) = \|\mathbf{A}_k\|_F^4 \cdot O_P\left(\frac{1}{T} \|\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j\|_F^4\right) \\
&= O_P\left(\frac{1}{T} \prod_{j=1}^K \|\mathbf{A}_j\|_F^4\right) = O_P\left(\frac{1}{T} \prod_{j=1}^K d_j^{2\alpha_{j,1}}\right), \tag{S.29}
\end{aligned}$$

and hence from (S.27), we have by Assumption (R1) that

$$|\tilde{R}(\gamma) - R^*(\gamma)| = O_P\left(\frac{1}{\sqrt{T}} d_k^{\alpha_{k,1} - \alpha_{k,r_k}}\right) = o_P(d^{-1/2} g_s^{1/2}) = o_P(1), \tag{S.30}$$

where the second last equality is from Assumption (R1). Next, with Proposition 1.3, consider

$$\begin{aligned}
\lambda_{r_k}(\mathbf{R}^*) &= \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Q} \mathbf{F}_{Z,t} \mathbf{\Lambda}' \mathbf{\Lambda} \mathbf{F}'_{Z,t} \mathbf{Q}' \right) = \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{A}_k \mathbf{F}_t \left[\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j \right]' \left[\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j \right] \mathbf{F}'_t \mathbf{A}_k' \right) \\
&\geq \lambda_{r_k}(\mathbf{A}_k' \mathbf{A}_k) \cdot \lambda_{r_k} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \left[\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j \right]' \left[\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j \right] \mathbf{F}'_t \right) \\
&\asymp_P d_k^{\alpha_{k,r_k}} \cdot \lambda_{r_k}(\text{tr}(\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j' \mathbf{A}_j) \Sigma_k) \asymp_P d_k^{\alpha_{k,r_k}} \prod_{j \in [K] \setminus \{k\}} d_j^{\alpha_{j,1}} = \omega_k.
\end{aligned}$$

With this, going back to the decomposition (S.20),

$$\omega_k^{-1} \lambda_{r_k}(\widehat{\mathbf{D}}) = \omega_k^{-1} \lambda_{r_k}(\widehat{\mathbf{S}}) \geq \omega_k^{-1} \lambda_{r_k}(\mathbf{R}^*) - \sup_{\|\gamma\|=1} |\widetilde{R}(\gamma) - R^*(\gamma)| - \sup_{\|\gamma\|=1} |R_1 + R_2 + R_3| \asymp_P 1,$$

where we used (S.25) and (S.30). Hence finally,

$$\|\widehat{\mathbf{D}}^{-1}\|_F = O_P(\lambda_{r_k}^{-1}(\widehat{\mathbf{D}})) = O_P(\omega_k^{-1}) = O_P\left(d_k^{\alpha_{k,1}-\alpha_{k,r_k}} \prod_{j=1}^K d_j^{-\alpha_{j,1}}\right),$$

which completes the proof of (S.18).

To prove (S.19), from (S.9) we obtain

$$\begin{aligned} \frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot} \right\|_F^2 &= \frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \widehat{\mathbf{D}}^{-1} (\mathcal{I}_j + \mathcal{I}\mathcal{I}_j + \mathcal{I}\mathcal{I}\mathcal{I}_j) \right\|_F^2 \leq \|\widehat{\mathbf{D}}^{-1}\|_F^2 \left[\frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \mathcal{I}_j + \mathcal{I}\mathcal{I}_j + \mathcal{I}\mathcal{I}\mathcal{I}_j \right\|_F^2 \right] \\ &\leq \|\widehat{\mathbf{D}}^{-1}\|_F^2 \left[\left(\frac{2}{d_k} \sum_{j=1}^{d_k} \|\mathcal{I}_j\|_F^2 \right) + \left(\frac{4}{d_k} \sum_{j=1}^{d_k} \|\mathcal{I}\mathcal{I}_j\|_F^2 \right) + \left(\frac{4}{d_k} \sum_{j=1}^{d_k} \|\mathcal{I}\mathcal{I}\mathcal{I}_j\|_F^2 \right) \right] \\ &= O_P\left(d_k^{2(\alpha_{k,1}-\alpha_{k,r_k})-1} \left(\frac{1}{Td_k} + \frac{1}{d_k} \right) \prod_{j=1}^K d_j^{2(1-\alpha_{j,1})}\right), \end{aligned}$$

where the last line follows from (S.18) and Lemma 1. This concludes the proof of Lemma 2. $\square$

Lemma 3 *Under the Assumptions in Lemma 2, for each $i, j \in [d_k]$ and $h \in [d_{-k}]$, if the set $\psi_{ij,h}$ has cardinality satisfying*

$$\frac{|\psi_{ij,h}|}{T} = 1 - \eta_{i,j,h} \geq 1 - \eta,$$

then for all $j \in [d_k]$ with $\mathbf{H}_j$ and $\mathbf{H}^a$ from (S.2),

$$\|\mathbf{H}_j - \mathbf{H}^a\|_F^2 = O_P\left(\min\left(\frac{1}{T}, \frac{\eta^2}{(1-\eta)^2}\right) d_k^{2(\alpha_{k,1}-\alpha_{k,r_k})}\right) = o_P(1). \quad (\text{S.31})$$

Proof of Lemma 3. Firstly, consider $\Delta_{F,k,ij,h}$ from (S.26), where

$$\begin{aligned} \|\Delta_{F,k,ij,h}\|_F &= \left\| \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{F}_t \mathbf{v}_h \mathbf{v}_h' \mathbf{F}_t' - \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{v}_h \mathbf{v}_h' \mathbf{F}_t' \right\|_F \\ &= \left\| \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}^c} \mathbf{F}_t \mathbf{v}_h \mathbf{v}_h' \mathbf{F}_t' \right\|_F + \left\| \left(\frac{1}{|\psi_{ij,h}|} - \frac{1}{T} \right) \sum_{t=1}^T \mathbf{F}_t \mathbf{v}_h \mathbf{v}_h' \mathbf{F}_t' \right\|_F \\ &\leq O_P\left(\frac{T\eta}{T-T\eta} \|\mathbf{v}_h\|^2\right) + \frac{T\eta}{T(T-T\eta)} \cdot O_P(T \|\mathbf{v}_h\|^2) \\ &= O_P\left(\frac{\eta}{1-\eta} \|\mathbf{v}_h\|^2\right). \end{aligned}$$

Combining this with (S.28), we have

$$\|\Delta_{F,k,ij,h}\|_F = O_P\left(\min\left(\frac{1}{\sqrt{T}}, \frac{\eta}{1-\eta}\right) \|\mathbf{v}_h\|^2\right). \quad (\text{S.32})$$

Note also that

$$\begin{aligned} \mathbf{H}_j &= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_{-k}} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \Lambda_{h\cdot}' \mathbf{F}_{Z,t}' \mathbf{Q}_{i\cdot} \Lambda_{h\cdot}' \mathbf{F}_{Z,t}' \\ &= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_{-k}} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_{i\cdot}' \mathbf{F}_{Z,t} \Lambda_{h\cdot} \Lambda_{h\cdot}' \mathbf{F}_{Z,t}', \end{aligned}$$

and also

$$\mathbf{H}^a = \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{T} \sum_{t=1}^T \mathbf{Q}'_{i\cdot} \mathbf{F}_{Z,t} \boldsymbol{\Lambda}_h \boldsymbol{\Lambda}'_h \mathbf{F}'_{Z,t}.$$

We then have, for $\widehat{\mathbf{A}}_{i\cdot} = \mathbf{Z}_k^{1/2} \widehat{\mathbf{Q}}_{i\cdot}$ and using (S.32),

$$\begin{aligned} \|\mathbf{H}_j - \mathbf{H}^a\|_F^2 &= \left\| \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \sum_{h=1}^{d_k} \left(\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{Q}'_{i\cdot} \mathbf{F}_{Z,t} \boldsymbol{\Lambda}_h \boldsymbol{\Lambda}'_h \mathbf{F}'_{Z,t} - \frac{1}{T} \sum_{t=1}^T \mathbf{Q}'_{i\cdot} \mathbf{F}_{Z,t} \boldsymbol{\Lambda}_h \boldsymbol{\Lambda}'_h \mathbf{F}'_{Z,t} \right) \right\|_F^2 \\ &= \left\| \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{i\cdot} \mathbf{A}'_{i\cdot} \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{Z}_k^{1/2} \right\|_F^2 \leq \|\widehat{\mathbf{D}}^{-1}\|_F^2 \left\| \sum_{i=1}^{d_k} \widehat{\mathbf{A}}_{i\cdot} \mathbf{A}'_{i\cdot} \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \right\|_F^2 \\ &= O_P(\omega_k^{-2}) \cdot \max_{i \in [d_k]} \left\| \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \right\|_F^2 \left(\sum_{i=1}^{d_k} \|\widehat{\mathbf{A}}_{i\cdot}\| \cdot \|\mathbf{A}_{i\cdot}\| \right)^2 \\ &= O_P(\omega_k^{-2}) \cdot O_P \left(\min \left(\frac{1}{T}, \frac{\eta^2}{(1-\eta)^2} \right) \|\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j\|_F^4 \right) \cdot \|\widehat{\mathbf{A}}_k\|_F^2 \cdot \|\mathbf{A}_k\|_F^2 \\ &= O_P \left(\min \left(\frac{1}{T}, \frac{\eta^2}{(1-\eta)^2} \right) d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} \right) = o_P(1), \end{aligned}$$

where we used (S.32) in the second last line, Assumption (L1) in the second last equality, and Assumption (R1) in the last equality. This completes the proof of Lemma 3. $\square$

Proof of Theorem 1.

The first result is shown in Lemma 2. Together with Lemma 3, we have

$$\begin{aligned} \frac{1}{d_k} \sum_{j=1}^{d_k} \|\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}^a \mathbf{Q}_{j\cdot}\|_F^2 &\leq \frac{1}{d_k} \sum_{j=1}^{d_k} \|\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot}\|_F^2 + \frac{1}{d_k} \sum_{j=1}^{d_k} \|(\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_{j\cdot}\|_F^2 \\ &\leq \frac{1}{d_k} \sum_{j=1}^{d_k} \|\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot}\|_F^2 + \frac{1}{d_k} \sum_{j=1}^{d_k} \|\mathbf{H}_j - \mathbf{H}^a\|_F^2 \|\mathbf{Q}_{j\cdot}\|_F^2 \\ &= O_P \left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) - 1} \left(\frac{1}{T d_k} + \frac{1}{d_k} \right) \prod_{j=1}^K d_j^{2(1 - \alpha_{j,1})} \right) \\ &\quad + O_P \left(\min \left(\frac{1}{T}, \frac{\eta^2}{(1-\eta)^2} \right) d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) - 1} \right) \\ &= O_P \left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) - 1} \left[\left(\frac{1}{T d_k} + \frac{1}{d_k} \right) \frac{d^2}{g_s^2} + \min \left(\frac{1}{T}, \frac{\eta^2}{(1-\eta)^2} \right) \right] \right), \end{aligned}$$

where we used $d_k^{-1} \sum_{j=1}^{d_k} \|\mathbf{Q}_{j\cdot}\|_F^2 = d_k^{-1} \|\mathbf{Q}_k\|_F^2 = O(d_k^{-1})$ by Assumption (L1). This completes the proof of the theorem. $\square$

Before we prove the consistency results for our imputations, we want to prove asymptotic normality for our factor loading estimators first. Consistency for the imputations will then use the rate obtained from asymptotic normality of the estimated factor loading matrices. We present a lemma first before proving Theorem 3.

Lemma 4 *Let Assumption (O1), (M1), (F1), (L1), (L2), (E1), (E2) and (R1) hold. For a given $k \in [K]$, let $\mathbf{R}^*$ be from (S.20) and $\omega_k := d_k^{\alpha_{k,r_k} - \alpha_{k,1}} g_s$. Then*

$$\begin{aligned} \omega_k^{-1} \mathbf{R}^* &\xrightarrow{P} \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \omega_k^{-1} \mathbf{A}_k \mathbf{A}'_k, \\ \omega_k^{-1} \widehat{\mathbf{D}}_k &\xrightarrow{P} \omega_k^{-1} \mathbf{D}_k := \omega_k^{-1} \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \text{diag}\{\lambda_j(\mathbf{A}'_k \mathbf{A}_k) \mid j \in [r_k]\}, \\ \mathbf{H}_k^a &\xrightarrow{P} \mathbf{H}_k^{a,*} := (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{1/2} \cdot \mathbf{D}_k^{-1/2} \Upsilon'_k \mathbf{Z}_k^{1/2}, \end{aligned}$$

where Υ_k is the eigenvector matrix of $\text{tr}(\mathbf{A}'_{-k}\mathbf{A}_{-k}) \cdot \omega_k^{-1}\mathbf{Z}_k^{1/2}\Sigma_{A,k}\mathbf{Z}_k^{1/2}$.

Proof of Lemma 4. First, let $\widehat{\mathbf{S}}, \widehat{\mathbf{Q}}, \widehat{\mathbf{D}}, \mathbf{H}^a$ be from (S.2), and $\mathbf{R}^*, \widetilde{\mathbf{R}}, \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$ from (S.20). Define also $\mathbf{H}^{a,*} := \mathbf{H}_k^{a,*}$, then we have

$$\begin{aligned} \frac{1}{\omega_k}(\widehat{\mathbf{D}} - \widehat{\mathbf{Q}}'\mathbf{R}^*\widehat{\mathbf{Q}}) &= \frac{1}{\omega_k}\widehat{\mathbf{Q}}'(\widehat{\mathbf{S}} - \mathbf{R}^*)\widehat{\mathbf{Q}} \\ &= \frac{1}{\omega_k}\widehat{\mathbf{Q}}'(\widetilde{\mathbf{R}} - \mathbf{R}^*)\widehat{\mathbf{Q}} + \frac{1}{\omega_k}\widehat{\mathbf{Q}}'\mathbf{R}_1\widehat{\mathbf{Q}} + \frac{1}{\omega_k}\widehat{\mathbf{Q}}'\mathbf{R}_2\widehat{\mathbf{Q}} + \frac{1}{\omega_k}\widehat{\mathbf{Q}}'\mathbf{R}_3\widehat{\mathbf{Q}} = o_P(1), \end{aligned} \quad (\text{S.33})$$

where the last equality follows from the proof of Lemma 2.

Using the structure in Assumption (F1), we have

$$\begin{aligned} \mathbb{E}\left[\frac{1}{\omega_k}\mathbf{R}^*\right] &= \frac{1}{\omega_k T} \sum_{t=1}^T \mathbb{E}\left[\mathbf{A}_k \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}_{f,t-q}\right) \mathbf{A}'_{-k} \mathbf{A}_{-k} \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q}\right) \mathbf{A}'_k\right] \\ &= \text{tr}(\mathbf{A}'_{-k}\mathbf{A}_{-k}) \cdot \frac{1}{\omega_k T} \sum_{t=1}^T \sum_{q \geq 0} a_{f,q}^2 \mathbf{A}_k \mathbf{A}'_k = \text{tr}(\mathbf{A}'_{-k}\mathbf{A}_{-k}) \cdot \frac{1}{\omega_k} \mathbf{A}_k \mathbf{A}'_k. \end{aligned}$$

Meanwhile, we have

$$\begin{aligned} \text{Var}\left(\left(\frac{1}{\omega_k}\mathbf{R}^*\right)_{ij}\right) &= \frac{1}{\omega_k^2 T^2} \sum_{t=1}^T \sum_{s=1}^T \text{cov}\left(\mathbf{A}'_{k,i} \cdot \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}_{f,t-q}\right) \mathbf{A}'_{-k} \mathbf{A}_{-k} \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q}\right) \mathbf{A}_{k,j}, \right. \\ &\quad \left. \mathbf{A}'_{k,i} \cdot \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}_{f,s-q}\right) \mathbf{A}'_{-k} \mathbf{A}_{-k} \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,s-q}\right) \mathbf{A}_{k,j}\right) \\ &= \frac{1}{\omega_k^2 T^2} \sum_{t=1}^T \sum_{q \geq 0} \sum_{p \geq 0} a_{f,q}^2 a_{f,p}^2 \cdot \text{Var}\left(\mathbf{A}'_{k,i} \cdot \mathbf{X}_{f,t-q} \mathbf{A}'_{-k} \mathbf{A}_{-k} \mathbf{X}'_{f,t-p} \mathbf{A}_{k,j}\right) \\ &= \frac{1}{\omega_k^2 T^2} \sum_{t=1}^T \sum_{q \geq 0} \sum_{p \geq 0} a_{f,q}^2 a_{f,p}^2 \cdot O_P(\|\mathbf{A}_{-k}\|_F^2) \\ &= O_P\left(d_k^{-2\alpha_{k,r_k}} \frac{1}{T} \prod_{j \in [K] \setminus \{k\}} d_j^{-\alpha_{j,1}}\right) = o_P(1), \end{aligned}$$

where we used Assumption (E2) in the third last equality, both (L1) and (F1) in the second last, and (R1) in the last. We can then conclude $\omega_k^{-1}\mathbf{R}^* \xrightarrow{P} \text{tr}(\mathbf{A}'_{-k}\mathbf{A}_{-k}) \cdot \omega_k^{-1}\mathbf{A}_k \mathbf{A}'_k$. Together with (S.33) and Assumption (L2), we obtain the limit of $\omega_k^{-1}\widehat{\mathbf{D}}$ as

$$\omega_k^{-1}\widehat{\mathbf{D}} \xrightarrow{P} \omega_k^{-1}\mathbf{D} = \omega_k^{-1}\text{tr}(\mathbf{A}'_{-k}\mathbf{A}_{-k}) \cdot \text{diag}\{\lambda_j(\mathbf{A}'_k \mathbf{A}_k) \mid j \in [r_k]\}. \quad (\text{S.34})$$

Define further

$$\mathbf{R}_{\text{res}} := \omega_k^{-1}\mathbf{Z}_k^{1/2}\mathbf{Q}'((\widetilde{\mathbf{R}} - \mathbf{R}^*) + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3)\widehat{\mathbf{Q}}.$$

With similar arguments in the proof of Lemma 2, we have $\|\mathbf{R}_{\text{res}}\|_F = o_P(\|\mathbf{Z}_k^{1/2}\|_F)$. Left-multiply both sides of $\widehat{\mathbf{S}}\widehat{\mathbf{Q}} = \widehat{\mathbf{Q}}\widehat{\mathbf{D}}$ by $\omega_k^{-1}\mathbf{Z}_k^{1/2}\mathbf{Q}'$, we can write

$$\begin{aligned} (\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}})(\omega_k^{-1}\widehat{\mathbf{D}}) &= \omega_k^{-1}\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{S}}\widehat{\mathbf{Q}} = \omega_k^{-1}\mathbf{Z}_k^{1/2}\mathbf{Q}'\mathbf{R}^*\widehat{\mathbf{Q}} + \mathbf{R}_{\text{res}} \\ &= \left[\omega_k^{-1}\mathbf{Z}_k^{1/2}\mathbf{Q}'\mathbf{R}^*\widehat{\mathbf{Q}}(\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}})^{-1} + \mathbf{R}_{\text{res}}(\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}})^{-1}\right](\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}}). \end{aligned}$$

Hence, each column of $\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}}$ is an eigenvector of the matrix

$$\omega_k^{-1}\mathbf{Z}_k^{1/2}\mathbf{Q}'\mathbf{R}^*\widehat{\mathbf{Q}}(\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}})^{-1} + \mathbf{R}_{\text{res}}(\mathbf{Z}_k^{1/2}\mathbf{Q}'\widehat{\mathbf{Q}})^{-1}.$$

We have $(\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}})' (\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}}) \xrightarrow{p} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1} \cdot \mathbf{D}$, since

$$\begin{aligned} & \omega_k^{-1} \left[(\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}})' (\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}}) - (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1} \cdot \mathbf{D} \right] \\ &= \left\{ \frac{1}{\omega_k} \widehat{\mathbf{Q}}' \mathbf{A}'_k \mathbf{A}_k \widehat{\mathbf{Q}} - \frac{1}{\omega_k} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1} \widehat{\mathbf{Q}}' \mathbf{R}^* \widehat{\mathbf{Q}} \right\} + \left\{ \frac{1}{\omega_k} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1} \widehat{\mathbf{Q}}' \mathbf{R}^* \widehat{\mathbf{Q}} - \frac{1}{\omega_k} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1} \mathbf{D} \right\}, \end{aligned}$$

which is $o_P(1)$ from the limit of $\omega_k^{-1} \mathbf{R}^*$ (for the first square bracket) and from (S.33) and (S.34) (for the second square bracket). Hence the eigenvalues of $(\mathbf{Q}' \widehat{\mathbf{Q}})' (\mathbf{Q}' \widehat{\mathbf{Q}})$ are asymptotically bounded away from zero and infinity by Assumption (L1), and also $\|(\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}})^{-1}\|_F = O_P(\|\mathbf{Z}_k^{-1/2}\|_F)$. Let

$$\Upsilon_k^* := (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{1/2} \cdot (\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}}) \mathbf{D}^{-1/2}.$$

Using the limit of $\omega_k^{-1} \mathbf{R}^*$, we have

$$\begin{aligned} \omega_k^{-1} \mathbf{Z}_k^{1/2} \mathbf{Q}' \mathbf{R}^* \widehat{\mathbf{Q}} (\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}})^{-1} & \xrightarrow{p} \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \omega_k^{-1} \mathbf{Z}_k^{1/2} \mathbf{Q}' \mathbf{Q} \mathbf{Z}_k \mathbf{Q}' \widehat{\mathbf{Q}} (\mathbf{Q}' \widehat{\mathbf{Q}})^{-1} \mathbf{Z}_k^{-1/2} \\ &= \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \omega_k^{-1} \mathbf{Z}_k^{1/2} \mathbf{Q}' \mathbf{Q} \mathbf{Z}_k^{1/2} \\ &= \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \omega_k^{-1} \mathbf{Z}_k^{1/2} \Sigma_{A,k} \mathbf{Z}_k^{1/2}, \end{aligned}$$

and $\|\mathbf{R}_{\text{res}}(\mathbf{Z}_k^{1/2} \mathbf{Q}' \widehat{\mathbf{Q}})^{-1}\|_F = o_P(1)$ from the above. By Assumption (L2) and eigenvector perturbation theories, there exists a unique eigenvector matrix Υ_k of $\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \omega_k^{-1} \mathbf{Z}_k^{1/2} \Sigma_{A,k} \mathbf{Z}_k^{1/2}$ such that $\|\Upsilon_k - \Upsilon_k^*\| = o_P(1)$. Therefore, we have

$$\mathbf{Q}' \widehat{\mathbf{Q}} = (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1/2} \cdot \mathbf{Z}_k^{-1/2} \Upsilon_k^* \mathbf{D}^{1/2} \xrightarrow{p} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{-1/2} \cdot \mathbf{Z}_k^{-1/2} \Upsilon_k \mathbf{D}^{1/2}.$$

Thus, we have

$$\begin{aligned} \mathbf{H}^a &= \widehat{\mathbf{D}}^{-1} \frac{1}{T} \sum_{t=1}^T \widehat{\mathbf{Q}}' \mathbf{A}_k \mathbf{F}_t \mathbf{A}'_{-k} \mathbf{A}_{-k} \mathbf{F}_t' \mathbf{Z}_k^{1/2} = \widehat{\mathbf{D}}^{-1} \widehat{\mathbf{Q}}' \mathbf{R}^* \mathbf{A}_k (\mathbf{A}'_k \mathbf{A}_k)^{-1} \mathbf{Z}_k^{1/2} = \widehat{\mathbf{D}}^{-1} \widehat{\mathbf{Q}}' \mathbf{R}^* \mathbf{Q} \Sigma_{A,k}^{-1} \\ &= \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot \mathbf{D}^{-1} \widehat{\mathbf{Q}}' \mathbf{Q} \mathbf{Z}_k \mathbf{Q}' \mathbf{Q} \Sigma_{A,k}^{-1} + o_P(1) \\ &\xrightarrow{p} (\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}))^{1/2} \cdot \mathbf{D}^{-1/2} \Upsilon_k \mathbf{Z}_k^{1/2}. \end{aligned} \tag{S.35}$$

This completes the proof of Lemma 4. $\square$

Proof of Theorem 3. Suppose we focus on the k -th mode, and hence we adapt all notations by omitting the subscript k for the ease of notational simplicity; see (S.2) for example. Moreover, we set $\mathbf{X}_{e,t} := \text{mat}_k(\mathcal{X}_{e,t})$, $\mathbf{X}_{\epsilon,t} := \text{mat}_k(\mathcal{X}_{\epsilon,t})$ and $\mathbf{X}_{f,t} := \text{mat}_k(\mathcal{X}_{f,t})$.

To proceed, we first decompose

$$\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}^a \mathbf{Q}_{j\cdot} = (\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot}) + (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_{j\cdot}. \tag{S.36}$$

Consider the first term $(\widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot})$. Using the decomposition in (S.9),

$$\begin{aligned} \widehat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot} &= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_{i\cdot} - \mathbf{H}_j \mathbf{Q}_{i\cdot}) \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}_t' \mathbf{A}_{i\cdot} \\ &\quad + \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_{i\cdot} - \mathbf{H}_j \mathbf{Q}_{i\cdot}) \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} (\mathbf{A}_{-k})'_h \mathbf{F}_t' \mathbf{A}_{j\cdot} \\ &\quad + \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_{i\cdot} - \mathbf{H}_j \mathbf{Q}_{i\cdot}) \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \\ &\quad + \mathcal{I}_{H,j} + \mathcal{I}_{H,j}^* + \mathcal{II}_{H,j} + \mathcal{III}_{H,j}, \quad \text{where} \end{aligned} \tag{S.37}$$

$$\begin{aligned}
\mathcal{I}_{H,j} &:= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \mathbf{H}^a \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i, \\
\mathcal{I}_{H,j}^* &:= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i, \\
\mathcal{II}_{H,j} &:= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_j, \\
\mathcal{III}_{H,j} &:= \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh}, \quad \text{with } \mathbf{A}_{-k} := \otimes_{l \in [K] \setminus \{k\}} \mathbf{A}_l.
\end{aligned}$$

We want to show that $\mathcal{I}_{H,j}$ is the leading term among those in (S.37). To this end, we will show $\sqrt{T\omega_B} \cdot \mathcal{I}_{H,j}$ converges to a normal distribution with mean zero and variance of constant order (see (S.47) later), so that $\mathcal{I}_{H,j}$ is of order $(T\omega_B)^{-1/2}$ exactly. Then it suffices to show that the rate $(T\omega_B)^{-1}$ is dominating the following rates multiplied by the rate of $\|\widehat{\mathbf{D}}^{-1}\|_F^2 = O_P\left(d_k^{2(\alpha_k, 1 - \alpha_k, r_k)} g_s^{-2}\right)$ from Lemma 2:

$$\left\| \sum_{i=1}^{d_k} (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \right\|^2, \quad (\text{S.38})$$

$$\left\| \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_j \right\|^2, \quad (\text{S.39})$$

$$\left\| \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right\|^2, \quad (\text{S.40})$$

$$\left\| \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_i - \mathbf{H}_j \mathbf{Q}_i) \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \right\|^2, \quad (\text{S.41})$$

$$\left\| \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_i - \mathbf{H}_j \mathbf{Q}_i) \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_j \right\|^2, \quad (\text{S.42})$$

$$\left\| \sum_{i=1}^{d_k} (\widehat{\mathbf{Q}}_i - \mathbf{H}_j \mathbf{Q}_i) \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right\|^2. \quad (\text{S.43})$$

Note that we can easily see the rates of (S.39) and (S.40) are greater than those of (S.42) and (S.43) respectively, using Lemma 2 and the Cauchy-Schwarz inequality.

Consider (S.38) first. We have

$$\begin{aligned}
& \left\| \sum_{i=1}^{d_k} (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \right\|^2 \\
& \leq \left(\sum_{i=1}^{d_k} \|\mathbf{H}_j - \mathbf{H}^a\|_F^2 \cdot \|\mathbf{Q}_i\|^2 \right) \cdot \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \right)^2 \\
& = O(d_k^2) \cdot \|\mathbf{H}_j - \mathbf{H}^a\|_F^2 \cdot \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \frac{1}{d_k \cdot |\psi_{ij,h}|} E_{t,jh} (\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \right)^2 \\
& = O(d_k^2) \cdot \|\mathbf{H}_j - \mathbf{H}^a\|_F^2 \cdot \sum_{i=1}^{d_k} \|\mathbf{u}_i\|^2 \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \frac{1}{d_k \cdot |\psi_{ij,h}|} E_{t,jh} \mathbf{v}'_h \mathbf{F}'_t \frac{1}{\|\mathbf{u}_i\|} \mathbf{u}_i \right)^2 \\
& = O_P \left(d_k^{2(\alpha_k, 1 - \alpha_k, r_k)} \left(\frac{d_k}{T^2} \right) d_k^{\alpha_k, 1} \right),
\end{aligned}$$

where $\mathbf{v}_h := (\mathbf{A}_{-k})_h$, $\mathbf{u}_i := \mathbf{A}_i$, and we used Lemma 3, Proposition 1.2 and (L1) in the last equality.

To bound (S.39), note from Assumption (E1) and (E2) that we can write

$$E_{t,ih} = \sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h} + [\text{mat}_k(\boldsymbol{\Sigma}_\epsilon)]_{ih} \sum_{q \geq 0} a_{e,q} (\mathbf{X}_{\epsilon,t-q})_{ih}.$$

Consider first $\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} (\sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h}) (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j$. By Assumption (O1), (E1), (E2) and (F1), we have

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \left(\sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h} \right) (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j \right)^2 \right] \\ &= \text{cov} \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} (\mathbf{A}_{-k})'_h \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q} \right) \mathbf{A}_j, \left(\sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h} \right), \right. \\ & \quad \left. \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} (\mathbf{A}_{-k})'_h \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q} \right) \mathbf{A}_j, \left(\sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h} \right) \right) \\ &= \sum_{h=1}^{d_k} \sum_{l=1}^{d_k} \sum_{t \in \psi_{ij,h} \cap \psi_{ij,l}} \sum_{q \geq 0} a_{f,q}^2 a_{e,q}^2 \cdot \|\mathbf{A}_j\|^2 \cdot \|(\mathbf{A}_{-k})_h\| \cdot \|(\mathbf{A}_{-k})_l\| \cdot \|\mathbf{A}_{e,-k,h}\| \cdot \|\mathbf{A}_{e,-k,l}\| \cdot \|\mathbf{A}_{e,k,i}\|^2 \\ &= O(T) \cdot \|\mathbf{A}_j\|^2 \cdot \|\mathbf{A}_{e,k,i}\|^2. \end{aligned} \tag{S.44}$$

Consider also $\sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_i \cdot ([\text{mat}_k(\boldsymbol{\Sigma}_\epsilon)]_{ih} \sum_{q \geq 0} a_{e,q} (\mathbf{X}_{\epsilon,t-q})_{ih}) (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j$. Similarly, by Assumption (O1), (E1), (E2) and (F1), we have

$$\begin{aligned} & \mathbb{E} \left[\left\| \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_i \cdot ([\text{mat}_k(\boldsymbol{\Sigma}_\epsilon)]_{ih} \sum_{q \geq 0} a_{e,q} (\mathbf{X}_{\epsilon,t-q})_{ih}) (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j \right\|^2 \right] \\ &= \text{cov} \left(\sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_i \cdot (\mathbf{A}_{-k})'_h \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q} \right) \mathbf{A}_j, ([\text{mat}_k(\boldsymbol{\Sigma}_\epsilon)]_{ih} \sum_{q \geq 0} a_{e,q} (\mathbf{X}_{\epsilon,t-q})_{ih}), \right. \\ & \quad \left. \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_i \cdot (\mathbf{A}_{-k})'_h \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q} \right) \mathbf{A}_j, ([\text{mat}_k(\boldsymbol{\Sigma}_\epsilon)]_{ih} \sum_{q \geq 0} a_{e,q} (\mathbf{X}_{\epsilon,t-q})_{ih}) \right) \\ &= \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \sum_{q \geq 0} a_{f,q}^2 a_{e,q}^2 \cdot \|\mathbf{A}_j\|^2 \cdot \|(\mathbf{A}_{-k})_h\|^2 \cdot \Sigma_{\epsilon,k,h,ii} \cdot \|\mathbf{Q}_i\|^2 \\ &= O(T) \cdot \|\mathbf{A}_j\|^2 \cdot \|\mathbf{A}_{-k}\|^2 \cdot \|\mathbf{Q}\|^2. \end{aligned} \tag{S.45}$$

Hence it holds that

$$\begin{aligned} & \left\| \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_i \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j \right\|^2 \\ & \leq \|\mathbf{H}_j\|_F^2 \cdot \left\| \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_i \cdot ([\text{mat}_k(\boldsymbol{\Sigma}_\epsilon)]_{ih} \sum_{q \geq 0} a_{e,q} (\mathbf{X}_{\epsilon,t-q})_{ih}) (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j \right\|^2 \\ & \quad + \|\mathbf{H}_j\|_F^2 \cdot \left(\sum_{i=1}^{d_k} \|\mathbf{Q}_i\|^2 \right) \cdot \sum_{i=1}^{d_k} \left\| \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \left(\sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h} \right) (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_j \right\|^2 \\ & = O_P \left(d_k^{-\alpha_{k,1}} \left(\frac{1}{T} \right) \prod_{j=1}^K d_j^{\alpha_{j,1}} \right), \end{aligned}$$

where we used Assumption (L1), (S.44) and (S.45) in the last equality.

For (S.40), by Assumption (O1), (E1) and (E2), we have from the proof of Proposition 1 that

$$\begin{aligned}
& \text{Var} \left(\sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \mathbf{Q}_{i \cdot} E_{t,ih} E_{t,jh} \right) \\
&= O(1) \cdot \sum_{i=1}^{d_k} \sum_{w=1}^{d_k} \sum_{h=1}^{d_k} \sum_{l=1}^{d_k} \sum_{t \in \psi_{ij,h} \cap \psi_{wj,l}} \sum_{n=1}^{r_{e,k}} \sum_{m=1}^{r_{e,-k}} \sum_{q \geq 0} a_{e,q}^4 A_{e,k,in} A_{e,k,wn} A_{e,k,jn}^2 A_{e,-k,hm}^2 A_{e,-k,lm}^2 \\
&\quad \cdot \|\mathbf{Q}_{i \cdot}\| \cdot \|\mathbf{Q}_{w \cdot}\| \cdot \text{Var}((\mathbf{X}_{e,t-q})_{nm}^2) \\
&+ O(1) \cdot \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h} \cap \psi_{wj,l}} \sum_{q \geq 0} a_{e,q}^4 \Sigma_{\epsilon,k,h,ii} \Sigma_{\epsilon,k,h,jj} \cdot \|\mathbf{Q}_{i \cdot}\|^2 \cdot \text{Var}((\mathbf{X}_{\epsilon,t-q})_{ih} (\mathbf{X}_{\epsilon,t-q})_{jh}) \\
&= O(T + T d_k) = O(T d_k).
\end{aligned}$$

Moreover, it holds that

$$\mathbb{E} \left[\sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right] = \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \left(\|\mathbf{A}_{e,-k,h \cdot}\|^2 \cdot \|\mathbf{A}_{e,k,i \cdot}\| \cdot \|\mathbf{A}_{e,k,j \cdot}\| + \Sigma_{\epsilon,k,h,ij} \right) = O(T d_k),$$

and with $\max_i \|\mathbf{Q}_{i \cdot}\|^2 \leq \|\mathbf{A}_j \cdot\|^2 \cdot \|\mathbf{Z}_k^{-1/2}\|^2 = O_P(d_k^{-\alpha_k, r_k})$, we thus have

$$\begin{aligned}
& \left\| \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_{i \cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right\|^2 \leq \|\mathbf{H}_j\|_F^2 \cdot \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{i \cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right\|^2 \\
&= O\left(\frac{d_k}{T} + d_k^2 d_k^{-\alpha_k, r_k}\right).
\end{aligned}$$

Now consider (S.41). Similar to (S.38), we have

$$\begin{aligned}
& \left\| \sum_{i=1}^{d_k} (\hat{\mathbf{Q}}_{i \cdot} - \mathbf{H}_j \mathbf{Q}_{i \cdot}) \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i \cdot} \right\|^2 \\
&\leq \left(\sum_{i=1}^{d_k} \|\hat{\mathbf{Q}}_{i \cdot} - \mathbf{H}_j \mathbf{Q}_{i \cdot}\|_F^2 \right) \cdot \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i \cdot} \right)^2 \\
&= O(d_k^2) \cdot \left(\sum_{i=1}^{d_k} \|\hat{\mathbf{Q}}_{i \cdot} - \mathbf{H}_j \mathbf{Q}_{i \cdot}\|_F^2 \right) \cdot \sum_{i=1}^{d_k} \|\mathbf{u}_i\|^2 \left(\sum_{h=1}^{d_k} \sum_{t \in \psi_{ij,h}} \frac{1}{d_k \cdot |\psi_{ij,h}|} E_{t,jh} \mathbf{v}'_h \mathbf{F}'_t \frac{1}{\|\mathbf{u}_i\|} \mathbf{u}_i \right)^2 \\
&= O_P \left(d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} \left(\frac{d_k}{T} \right) \left(\frac{1}{T d_k} + \frac{1}{d_k} \right) \prod_{j=1}^K d_j^{2(1-\alpha_{j,1})} \right),
\end{aligned}$$

where we used Lemma 2, Proposition 1.2 and (L1) in the last equality.

Finally, we consider the following ratios with $d_1, \dots, d_K, T \rightarrow \infty$:

$$\begin{aligned}
& \left\| \sum_{i=1}^{d_k} (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \right\|^2 d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} g_s^{-2} / (T\omega_B)^{-1} \\
&= O_P \left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} \cdot \frac{1}{T} \right) = o_P(1), \\
& \left\| \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{j\cdot} \right\|^2 d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} g_s^{-2} / (T\omega_B)^{-1} \\
&= O_P \left(d_k^{-\alpha_{k,1}} \prod_{j \in [K] \setminus \{k\}} d_j^{\alpha_{j,1} - 1} \right) = o_P(1), \\
& \left\| \sum_{i=1}^{d_k} \mathbf{H}_j \mathbf{Q}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,ih} E_{t,jh} \right\|^2 d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} g_s^{-2} / (T\omega_B)^{-1} \\
&= O_P \left(T d_{-k} d_k^{-\alpha_{k,r_k} - \alpha_{k,1}} \right) = o_P(1), \\
& \left\| \sum_{i=1}^{d_k} (\hat{\mathbf{Q}}_{i\cdot} - \mathbf{H}_j \mathbf{Q}_{i\cdot}) \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \right\|^2 d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} g_s^{-2} / (T\omega_B)^{-1} \\
&= O_P \left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} \left(\frac{1}{T d_{-k}} + \frac{1}{d_k} \right) \prod_{j=1}^K d_j^{2(1 - \alpha_{j,1})} \right) = o_P(1),
\end{aligned}$$

by Assumptions (R1) and the rate assumptions $d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k}} = o(d_{-k})$ (from the statement of Theorem 2) and $T d_{-k} = o(d_k^{\alpha_{k,1} + \alpha_{k,r_k}})$. Hence $\mathcal{I}_{H,j}$ is indeed the dominating term in (S.37). In other words, we have

$$\hat{\mathbf{Q}}_{j\cdot} - \mathbf{H}_j \mathbf{Q}_{j\cdot} = \mathcal{I}_{H,j} + o_P(1). \quad (\text{S.46})$$

Then we want to show that

$$\begin{aligned}
\sqrt{T\omega_B} \cdot \mathcal{I}_{H,j} &= \sqrt{T\omega_B} \cdot \hat{\mathbf{D}}^{-1} \mathbf{H}^a \sum_{i=1}^{d_k} \mathbf{Q}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \\
&\xrightarrow{P} \sqrt{T\omega_B} \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \sum_{i=1}^{d_k} \mathbf{Q}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \\
&\xrightarrow{D} \mathcal{N}(\mathbf{0}, T\omega_B \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \boldsymbol{\Xi}_{k,j} (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1}),
\end{aligned} \quad (\text{S.47})$$

where $\mathbf{D}$ and $\mathbf{H}_k^{a,*}$ are from Lemma 4, and we require Assumption (AD1) for the covariance matrix to have constant rate. In fact, using Lemma 4 and Proposition 1, the upper bound is of constant order by

$$\begin{aligned}
& \left\| T\omega_B \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \boldsymbol{\Xi}_{k,j} (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1} \right\|_F \\
&= O(T\omega_B) \cdot \|\mathbf{D}^{-1}\|_F^2 \cdot \left\| \sum_{i=1}^{d_k} \mathbf{Q}_{i\cdot} \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \right\|^2 \\
&= O \left(T (d_{-k} d_k^{\alpha_{k,1}})^{-1} \right) \cdot \left(\sum_{i=1}^{d_k} \|\mathbf{Q}_{i\cdot}\|^2 \right) \cdot \sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh} (\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{i\cdot} \right)^2 = O_P(1).
\end{aligned}$$

We will adapt the central limit theorem for α -mixing processes (Fan and Yao (2003), Theorem 2.21). Due to the existence of missing data and the general missing patterns that we allow, we construct an

auxiliary time series to facilitate the proof. Formally, define $\{\mathbf{B}_{j,t}\}_{t \in [T]}$ as

$$\mathbf{B}_{j,t} := \sqrt{\omega_B} \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \mathbf{Q}_i \cdot \frac{T}{|\psi_{ij,h}|} \cdot E_{t,jh}(\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \cdot \mathbb{1}\{t \in \psi_{ij,h}\}. \quad (\text{S.48})$$

Hence we have the following,

$$\sqrt{T\omega_B} \cdot \mathcal{I}_{H,j} \xrightarrow{p} \frac{1}{\sqrt{T}} \sum_{t=1}^T \mathbf{B}_{j,t}.$$

It is easy to see that $\mathbb{E}[\mathbf{B}_{j,t}] = \mathbf{0}$ by Assumption (E1), (E2) and (F1). Moreover, $\mathbf{B}_{j,t}$ is also α -mixing over t . To see this, consider

$$E_{t,jh}(\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i = \left\{ \sum_{q \geq 0} a_{e,q} \mathbf{A}'_{e,k,i} \cdot \mathbf{X}_{e,t-q} \mathbf{A}_{e,-k,h} + \sum_{\epsilon,k,h,ii}^{1/2} \sum_{q \geq 0} a_{\epsilon,q} (\mathbf{X}_{\epsilon,t-q})_{ih} \right\} \mathbf{A}'_{-k,h} \cdot \left(\sum_{q \geq 0} a_{f,q} \mathbf{X}'_{f,t-q} \right) \mathbf{A}_i.$$

If we define $\mathbf{b}_{e,t} := \sum_{q \geq 0} a_{e,q} \mathbf{X}_{e,t-q}$, $\mathbf{b}_{\epsilon,ih,t} := \sum_{q \geq 0} a_{\epsilon,q} (\mathbf{X}_{\epsilon,t-q})_{ih}$ and $\mathbf{b}_{f,t} := \sum_{q \geq 0} a_{f,q} \mathbf{X}_{f,t-q}$ which are independent of each other by Assumption (E2), we can then rewrite

$$\mathbf{B}_{j,t} = h\left(\mathbf{b}_{e,t}, (\mathbf{b}_{\epsilon,ih,t})_{i \in [d_k], h \in [d_k]}, \mathbf{b}_{f,t}\right),$$

for some function h , and hence Theorem 5.2 in Bradley (2005) implies the α -mixing property. Then similar to Chen and Fan (2023), it is left to show that there exists an $m > 2$ such that $\mathbb{E}[\|\mathbf{B}_{j,t}\|^m] \leq C$ for some constant C . With Assumption (E1), (E2) and (F1) and similar to the proof of Proposition 1, we have

$$\mathbb{E} \left[\sum_{h=1}^{d_k} \frac{T}{|\psi_{ij,h}|} \cdot E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v} \cdot \mathbb{1}\{t \in \psi_{ij,h}\} \right]^2 = O(d_k),$$

where $\mathbf{u} \in \mathbb{R}^{r_k}$ and $\mathbf{v} \in \mathbb{R}^{r_k}$ are any deterministic vectors of constant order. Hence

$$\begin{aligned} & \mathbb{E}[\|\mathbf{B}_{j,t}\|^m] \\ & \leq \omega_B^{m/2} \|\mathbf{D}^{-1}\|_F^m \|\mathbf{H}_k^{a,*}\|_F^m \left(\sum_{i=1}^{d_k} \|\mathbf{Q}_i\|^2 \right)^{m/2} \mathbb{E} \left\{ \left[\sum_{i=1}^{d_k} \left(\sum_{h=1}^{d_k} \frac{T}{|\psi_{ij,h}|} E_{t,jh}(\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \cdot \mathbb{1}\{t \in \psi_{ij,h}\} \right)^2 \right]^{m/2} \right\} \\ & = O\left((\omega_B d_k d_k^{\alpha_{k,1}})^{m/2}\right) \cdot \|\mathbf{D}^{-1}\|_F^m = O_P \left(\left(\omega_B d_k d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} \prod_{j=1}^K d_j^{-2\alpha_{j,1}} \right)^{m/2} \right) = O_P(1), \end{aligned}$$

where we used Lemma 2 and the definition of ω_B in the last line. Theorem 2.21 in Fan and Yao (2003) then applies. With (S.46), (S.47) and Lemma 4, we can directly establish that

$$\sqrt{T\omega_B} \cdot (\widehat{\mathbf{Q}}_j - \mathbf{H}_j \mathbf{Q}_j) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T\omega_B \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \mathbf{\Xi}_{k,j} (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1}). \quad (\text{S.49})$$

Consider now the second term in (S.36). By Lemma 3 and 4, we have

$$\left\| (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_j \right\|^2 \leq \|\mathbf{H}_j - \mathbf{H}^a\|_F^2 \cdot \|\mathbf{A}_j\|^2 \cdot \|\mathbf{Z}_k^{-1/2}\|^2 = O_P \left(\min \left(\frac{1}{T}, \frac{\eta^2}{(1-\eta)^2} \right) d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k}} \right),$$

implying

$$\begin{aligned} & \left\| (\mathbf{H}_j - \mathbf{H}^a) \mathbf{Q}_j \right\|^2 \left/ \left\| \widehat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \mathbf{H}^a \mathbf{Q}_i \sum_{h=1}^{d_k} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} E_{t,jh}(\mathbf{A}_{-k})'_h \cdot \mathbf{F}'_t \mathbf{A}_i \right\|^2 \right. \\ & = O_P \left(\min \left(1, \frac{T\eta^2}{(1-\eta)^2} \right) d_k^{3(\alpha_{k,1} - \alpha_{k,r_k})} \prod_{j \in [K] \setminus \{k\}} d_j^{2\alpha_{j,1} - 1} \right), \end{aligned}$$

which is unrealistic to be $o_P(1)$ in the presence of missing data in general. Thus $(\mathbf{H}_j - \mathbf{H}^a)\mathbf{Q}_j$ contributes to the asymptotic distribution of $(\hat{\mathbf{Q}}_j - \mathbf{H}^a\mathbf{Q}_j)$. Rewrite

$$\begin{aligned} (\mathbf{H}_j - \mathbf{H}^a)\mathbf{Q}_j &= \hat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_i \cdot \sum_{h=1}^{d_k} \left(\frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \mathbf{Q}'_i \mathbf{F}_{Z,t} \mathbf{\Lambda}_h \mathbf{\Lambda}'_h \mathbf{F}'_{Z,t} - \frac{1}{T} \sum_{t=1}^T \mathbf{Q}'_i \mathbf{F}_{Z,t} \mathbf{\Lambda}_h \mathbf{\Lambda}'_h \mathbf{F}'_{Z,t} \right) \mathbf{Q}_j \\ &= \hat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} \hat{\mathbf{Q}}_i \cdot \mathbf{A}'_i \cdot \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{Z}_k^{1/2} \mathbf{Q}_j \\ &= \hat{\mathbf{D}}^{-1} \sum_{i=1}^{d_k} (\hat{\mathbf{Q}}_i - \mathbf{H}^a \mathbf{Q}_i) \mathbf{A}'_i \cdot \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{Z}_k^{1/2} \mathbf{Q}_j + \hat{\mathbf{D}}^{-1} \mathbf{H}^a \sum_{i=1}^{d_k} \mathbf{Q}_i \cdot \mathbf{A}'_i \cdot \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{Z}_k^{1/2} \mathbf{Q}_j. \end{aligned}$$

Note the first term is dominated by the second term due to Theorem 1. Using Assumption (AD2) and the Slutsky's theorem, we have

$$\sqrt{T d_k^{\alpha_k, r_k}} \cdot \hat{\mathbf{D}}^{-1} \mathbf{H}^a \sum_{i=1}^{d_k} \mathbf{Q}_i \cdot \mathbf{A}'_i \cdot \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{Z}_k^{1/2} \mathbf{Q}_j \rightarrow \mathcal{N}(\mathbf{0}, \mathbf{D}^{-1} \mathbf{H}_k^{a,*} h_{k,j}(\mathbf{A}_j) (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1}) \quad \mathcal{G}^T\text{-stably.} \quad (\text{S.50})$$

Furthermore, $\mathcal{I}_{H,j}$ and $(\mathbf{H}_j - \mathbf{H}^a)\mathbf{Q}_j$ are asymptotically independent since the randomness of $\mathcal{I}_{H,j}$ comes from $E_{t,jh}(\mathbf{A}_{-k})'_h \mathbf{F}'_t$ while that of $(\mathbf{H}_j - \mathbf{H}^a)\mathbf{Q}_j$ comes from $\Delta_{F,k,ij,h}$. From (S.49) and (S.50), we conclude that

$$\sqrt{T d_k^{\alpha_k, r_k}} \cdot (\hat{\mathbf{Q}}_j - \mathbf{H}^a \mathbf{Q}_j) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{D}^{-1} \mathbf{H}_k^{a,*} (T d_k^{\alpha_k, r_k} \cdot \mathbf{\Xi}_{k,j} + h_{k,j}(\mathbf{A}_j)) (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1}).$$

On the other hand, if we have finite missingness or asymptotically vanishing missingness such that

$$\min \left(1, \frac{T\eta^2}{(1-\eta)^2} \right) d_k^{3(\alpha_{k,1} - \alpha_{k, r_k})} \prod_{j \in [K] \setminus \{k\}} d_j^{2\alpha_{j,1} - 1} = T d^{-1} g_s^2 g_\eta d_k^{1 + \alpha_{k,1} - 3\alpha_{k, r_k}} = o(1),$$

then (S.50) is dominated by (S.49), and hence it holds at the absence of (AD2) that

$$\sqrt{T \omega_B} \cdot (\hat{\mathbf{Q}}_j - \mathbf{H}^a \mathbf{Q}_j) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T \omega_B \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \mathbf{\Xi}_{k,j} (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1}).$$

This completes the proof of Theorem 3. $\square$

Proof of Theorem 4. By Lemma 4, $\hat{\mathbf{D}}_k$ is consistent for $\mathbf{D}_k$, and $\mathbf{H}_k^a$ is consistent for $\mathbf{H}_k^{a,*}$. Similar to the proof of Theorem 5 in Chen and Fan (2023), it suffice to prove that the HAC estimator $\hat{\mathbf{\Sigma}}_{HAC}$ based on $\{\hat{\mathbf{Q}}_k, \text{mat}_k(\hat{\mathcal{C}}_t), \text{mat}_k(\hat{\mathcal{E}}_t)\}_{t \in [T]}$ is a consistent estimator for $\mathbf{H}_k^a \mathbf{\Xi}_{k,j} (\mathbf{H}_k^a)'$. Recall that

$$\begin{aligned} \mathbf{H}_k^a \mathbf{\Xi}_{k,j} (\mathbf{H}_k^a)' &= \text{Var} \left(\sum_{i=1}^{d_k} \mathbf{H}_k^a \mathbf{Q}_{k,i} \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} E_{t,jh}(\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{k,i} \right) \\ &= \text{Var} \left(\sum_{i=1}^{d_k} \left(\hat{\mathbf{D}}_k^{-1} \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{Q}}_k' \mathbf{Q}_k \mathbf{F}_{Z,t} \mathbf{\Lambda}'_t \mathbf{\Lambda} \mathbf{F}'_{Z,t} \right) \mathbf{Q}_{k,i} \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} E_{t,jh}(\mathbf{A}_{-k})'_h \mathbf{F}'_t \mathbf{A}_{k,i} \right) \\ &= \text{Var} \left(\sum_{i=1}^{d_k} \left(\frac{1}{T} \sum_{t=1}^T \hat{\mathbf{D}}_k^{-1} \hat{\mathbf{Q}}_k' \cdot \text{mat}_k(\mathcal{C}_t) \cdot \text{mat}_k(\mathcal{C}_t)_i \right) \right. \\ &\quad \left. \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} \text{mat}_k(\mathcal{E}_t)_{jh} \cdot \text{mat}_k(\mathcal{C}_t)_{ih} \right). \end{aligned}$$

By Theorem 1, and the rate assumption in the statement of Theorem 4, we have $\hat{\mathbf{Q}}_k$ being consistent for a version of $\mathbf{Q}_k$ (in Frobenius norm) for any $k \in [K]$. By Theorem 2 and the assumption that the

rate for individual common component imputation error is going to 0, $\widehat{\mathcal{C}}_{t,i_1,\dots,i_K}$ is consistent for $\mathcal{C}_{t,i_1,\dots,i_K}$ for any $k \in [K], i_k \in [d_k], t \in [T]$. Hence, it also holds that $\widehat{\mathcal{E}}_{t,i_1,\dots,i_K}$ is consistent for $\mathcal{E}_{t,i_1,\dots,i_K}$ for any $k \in [K], i_k \in [d_k], t \in [T]$. We can finally conclude that $\widehat{\Sigma}_{HAC}$ is estimating $\mathbf{H}_k^a \Xi_{k,j} (\mathbf{H}_k^a)'$ consistently (Newey and West, 1987), which is result 1. We can also show a similar result for $\widehat{\Sigma}_{HAC}^\Delta$, which is result 2 (details omitted). Combining both results, and consider the general statement of Theorem 3, we can easily conclude result 3. This completes the proof of the theorem. $\square$

We will present two other lemmas before proving Theorem 2. While we stick with the notations in (S.2), we use the following also hereafter:

$$\begin{aligned} \mathbf{y}_t &:= \text{vec}(\mathcal{Y}_t), \mathbf{m}_t := \text{vec}(\mathcal{M}_t), \mathbf{f}_{Z,t} := \text{vec}(\mathcal{F}_{Z,t}), \boldsymbol{\varepsilon}_t := \text{vec}(\mathcal{E}_t), \mathbf{c}_t := \text{vec}(\mathcal{C}_t), \mathbf{f}_t := \text{vec}(\mathcal{F}_t), \\ \mathbf{H}_\otimes &:= \mathbf{H}_K^a \otimes \dots \otimes \mathbf{H}_1^a, \mathbf{A}_\otimes := \mathbf{A}_K \otimes \dots \otimes \mathbf{A}_1, \mathbf{Z}_\otimes := \mathbf{Z}_K \otimes \dots \otimes \mathbf{Z}_1, \end{aligned} \quad (\text{S.51})$$

where the hat versions (if any) of the above are defined similarly.

Lemma 5 *Under the assumptions in Theorem 2, for any $k \in [K]$ and $j \in [d_k]$,*

$$\|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k^a \mathbf{Q}_{k,j\cdot}\|_F^2 = O_P\left(T^{-1} d_{-k} d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-2} + d^2 g_s^{-2} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 2} + g_\eta d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k}}\right). \quad (\text{S.52})$$

Proof of Lemma 5. First, consider the case when $Td_{-k} = o(d_k^{\alpha_{k,r_k} + \alpha_{k,1}})$. From (S.49) in the proof of Theorem 3, we have $\|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_{k,j} \mathbf{Q}_{k,j\cdot}\|_F^2 = O_P(T^{-1} \omega_B^{-1})$. Hence it follows that

$$\begin{aligned} \|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k^a \mathbf{Q}_{k,j\cdot}\|_F^2 &= O_P\left(\|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_{k,j} \mathbf{Q}_{k,j\cdot}\|_F^2 + \|(\mathbf{H}_{k,j} - \mathbf{H}_k^a) \mathbf{Q}_{k,j\cdot}\|_F^2\right) \\ &= O_P\left((T\omega_B)^{-1} + g_\eta d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k}}\right) = O_P\left(T^{-1} d_{-k} d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-2} + g_\eta d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k}}\right), \end{aligned} \quad (\text{S.53})$$

where we used Lemma 3 in the second equality, and

$$\|\mathbf{Q}_{k,j\cdot}\|^2 = \|\mathbf{Z}_k^{-1/2} \mathbf{A}_{k,j\cdot}\|^2 = O_P(d_k^{-\alpha_{k,r_k}}).$$

Now suppose $Td_{-k} = o(d_k^{\alpha_{k,r_k} + \alpha_{k,1}})$ fails to hold. From the decomposition of $\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_{k,j} \mathbf{Q}_{k,j\cdot}$ in (S.37), $\mathcal{I}_{H,j}$ is not the leading term anymore, and the leading term among the expressions from (S.38) to (S.43) will be (S.40). It has rate

$$O_P\left(\frac{d_{-k}}{T} + d_{-k}^2 d_k^{-\alpha_{k,r_k}}\right) = O_P(d_{-k}^2 d_k^{-\alpha_{k,r_k}}),$$

where the above equality used the fact that $Td_{-k} = o(d_k^{\alpha_{k,r_k} + \alpha_{k,1}})$ does not hold. Together with the bound on $\|\widehat{\mathbf{D}}_k^{-1}\|_F$ from Lemma 2, we have

$$\|\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_{k,j} \mathbf{Q}_{k,j\cdot}\|_F^2 = O_P\left(d^2 g_s^{-2} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 2}\right). \quad (\text{S.54})$$

Combining (S.53) and (S.54), we arrive at the statement of the lemma. $\square$

Lemma 6 *Under the Assumptions in Theorem 2, with the notations in (S.2) and (S.51), we have the following for any $j \in [d]$:*

$$\|\mathbf{Q}_\otimes \mathbf{H}'_\otimes\|_F^2 = O_P(1), \quad (\text{S.55})$$

$$\begin{aligned} &\|\widehat{\mathbf{Q}}_{\otimes,j\cdot} - \mathbf{H}_\otimes \mathbf{Q}_{\otimes,j\cdot}\|^2 \\ &= O_P\left(\max_{k \in [K]} \left\{ T^{-1} d_{-k} d_k^{3\alpha_{k,1} - \alpha_{k,r_k}} g_s^{-2} g_w^{-1} + d^2 g_s^{-2} g_w^{-1} d_k^{2\alpha_{k,1} - 2\alpha_{k,r_k} - 2} + g_\eta g_w^{-1} d_k^{2\alpha_{k,1} - 2\alpha_{k,r_k}} \right\}\right), \end{aligned} \quad (\text{S.56})$$

$$\begin{aligned} &\|\widehat{\mathbf{Q}}_\otimes - \mathbf{Q}_\otimes \mathbf{H}'_\otimes\|_F^2 \\ &= O_P\left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-2} + d^2 g_s^{-2} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 1} + g_\eta d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} + 1} \right\}\right). \end{aligned} \quad (\text{S.57})$$

Proof of Lemma 6. For (S.55), we have

$$\|\mathbf{Q}_{\otimes} \mathbf{H}'_{\otimes}\|_F^2 \leq \|\mathbf{H}_{\otimes}\|_F^2 \cdot \prod_{k=1}^K \|\mathbf{Q}_k\|_F^2 = O_P(1),$$

where we used Assumption (L1) in the last equality.

To show (S.56), for any $j \in [d_k]$, by a simple induction argument (omitted),

$$\begin{aligned} \|\widehat{\mathbf{Q}}_{\otimes, j} - \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes, j}\|^2 &= \|(\widehat{\mathbf{Q}}_{\otimes} - \mathbf{Q}_{\otimes} \mathbf{H}'_{\otimes})_j\|^2 = \left\| [(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}'_K \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}'_1)]_j \right\|^2 \\ &\leq \sum_{k=1}^K \left\{ \max_{j \in [d_k]} \|\widehat{\mathbf{Q}}_{k, j} - \mathbf{H}_k^a \mathbf{Q}_{k, j}\|^2 \prod_{\ell \in [K] \setminus \{k\}} \max_{j \in [d_{\ell}]} \|\widehat{\mathbf{Q}}_{\ell, j}\|^2 \right\} \\ &= O_P \left(\max_{k \in [K]} \left\{ \left[T^{-1} d_k d_k^{3\alpha_{k,1} - 2\alpha_{k, r_k}} g_s^{-2} + d^2 g_s^{-2} d_k^{2\alpha_{k,1} - 3\alpha_{k, r_k} - 2} + g_{\eta} d_k^{2\alpha_{k,1} - 3\alpha_{k, r_k}} \right] \cdot \prod_{\ell \in [K] \setminus \{k\}} d_{\ell}^{-\alpha_{\ell, r_{\ell}}} \right\} \right) \\ &= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d_k d_k^{3\alpha_{k,1} - \alpha_{k, r_k}} g_s^{-2} g_w^{-1} + d^2 g_s^{-2} g_w^{-1} d_k^{2\alpha_{k,1} - 2\alpha_{k, r_k} - 2} + g_{\eta} g_w^{-1} d_k^{2\alpha_{k,1} - 2\alpha_{k, r_k}} \right\} \right), \end{aligned}$$

where the second last equality used (S.52) and

$$\begin{aligned} \|\widehat{\mathbf{Q}}_{\ell, j}\|^2 &\leq 2(\|\widehat{\mathbf{Q}}_{\ell, j} - \mathbf{H}_{\ell}^a \mathbf{Q}_{\ell, j}\|^2 + \|\mathbf{H}_{\ell}^a \mathbf{Q}_{\ell, j}\|^2) = O_P(\|\widehat{\mathbf{Q}}_{\ell, j} - \mathbf{H}_{\ell}^a \mathbf{Q}_{\ell, j}\|^2 + \|\mathbf{H}_{\ell}^a \mathbf{Z}_{\ell}^{-1/2} \mathbf{A}_{\ell, j}\|^2) \\ &= O_P(\|\widehat{\mathbf{Q}}_{\ell, j} - \mathbf{H}_{\ell}^a \mathbf{Q}_{\ell, j}\|^2 + \|\mathbf{H}_{\ell}^a\|_F^2 \cdot (d_{\ell}^{-\alpha_{\ell, r_{\ell}}/2})^2 \cdot 1) = O_P(d_{\ell}^{-\alpha_{\ell, r_{\ell}}}). \end{aligned}$$

Finally it also holds that

$$\begin{aligned} \|\widehat{\mathbf{Q}}_{\otimes} - \mathbf{Q}_{\otimes} \mathbf{H}'_{\otimes}\|_F^2 &= \|(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_1) - (\mathbf{Q}_K \mathbf{H}_K^{a'} \otimes \cdots \otimes \mathbf{Q}_1 \mathbf{H}_1^{a'})\|_F^2 = O(1) \cdot \sum_{k=1}^K \|\widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k^{a'}\|_F^2 \\ &= O \left(\max_{k \in [K]} \sum_{j=1}^{d_k} \|\widehat{\mathbf{Q}}_{k, j} - \mathbf{H}_k^a \mathbf{Q}_{k, j}\|^2 \right) \\ &= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1} - 2\alpha_{k, r_k}} g_s^{-2} + d^2 g_s^{-2} d_k^{2\alpha_{k,1} - 3\alpha_{k, r_k} - 1} + g_{\eta} d_k^{2\alpha_{k,1} - 3\alpha_{k, r_k} + 1} \right\} \right), \end{aligned}$$

where the second equality could be shown by a simple induction argument using $\|\mathbf{Q}_k\| = O(1)$ (omitted), and the last equality is from (S.52). $\square$

Proof of Theorem 2. The equation (3.7) is essentially

$$\begin{aligned}
\hat{\mathbf{f}}_{Z,t} &= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} y_{t,j} \right) \\
&= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} (\mathbf{Q}'_{\otimes,j} \mathbf{f}_{Z,t} + \varepsilon_{t,j}) \right) \\
&= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \mathbf{Q}'_{\otimes,j} \mathbf{f}_{Z,t} \right) + \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \varepsilon_{t,j} \right) \\
&= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right) (\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t} \\
&\quad + \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \varepsilon_{t,j} \right) \\
&\quad + \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} (\mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} - \hat{\mathbf{Q}}_{\otimes,j})' \right) (\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t} \\
&=: (\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t} + \tilde{\varepsilon}_{H,t} + \tilde{\varepsilon}_t + \tilde{\mathbf{f}}_{Z,t}, \quad \text{where}
\end{aligned} \tag{S.58}$$

$$\begin{aligned}
\tilde{\varepsilon}_{H,t} &:= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \varepsilon_{t,j} \right), \\
\tilde{\varepsilon}_t &:= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} (\hat{\mathbf{Q}}_{\otimes,j} - \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j}) \varepsilon_{t,j} \right), \\
\tilde{\mathbf{f}}_{Z,t} &:= \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \right)^{-1} \left(\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} (\mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} - \hat{\mathbf{Q}}_{\otimes,j})' \right) (\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t}.
\end{aligned}$$

Then we have

$$\begin{aligned}
&\left\| \sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} - \sum_{j=1}^d m_{t,j} \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \mathbf{Q}'_{\otimes,j} \mathbf{H}'_{\otimes} \right\|_F \\
&\leq \sum_{j=1}^d \left\| \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} - \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \mathbf{Q}'_{\otimes,j} \mathbf{H}'_{\otimes} \right\|_F \\
&\leq \sum_{j=1}^d \left\| \hat{\mathbf{Q}}_{\otimes,j} - \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \right\|^2 + 2 \sum_{j=1}^d \left\| \hat{\mathbf{Q}}_{\otimes,j} - \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \right\| \cdot \left\| \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \right\| \\
&= O_P(\left\| \hat{\mathbf{Q}}_{\otimes} - \mathbf{Q}_{\otimes} \mathbf{H}'_{\otimes} \right\|_F^2 + \left\| \hat{\mathbf{Q}}_{\otimes} - \mathbf{Q}_{\otimes} \mathbf{H}'_{\otimes} \right\|_F) \\
&= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-2} + d^2 g_s^{-2} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 1} + g_{\eta} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} + 1} \right\}^{1/2} \right) = o_P(1),
\end{aligned}$$

where we used the Cauchy-Schwarz inequality, (S.55) and (S.57) in the last equality, and Assumption (R1).

Hence we have

$$\sum_{j=1}^d m_{t,j} \hat{\mathbf{Q}}_{\otimes,j} \hat{\mathbf{Q}}'_{\otimes,j} \xrightarrow{p} \sum_{j=1}^d m_{t,j} \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \mathbf{Q}'_{\otimes,j} \mathbf{H}'_{\otimes}. \tag{S.59}$$

Note that by (S.35) we have $\left\| \mathbf{H}_{\otimes}^{-1} \right\|_F = O_P(1) \cdot \left\| \mathbf{Q}'_K \hat{\mathbf{Q}}_K \otimes \cdots \otimes \mathbf{Q}'_1 \hat{\mathbf{Q}}_1 \right\| = O_P(1)$, which will be used later

in the proof. To bound $\mathbf{f}_{Z,t}$, from Assumption (F1), we have

$$\mathbb{E}\|\mathbf{f}_t\|^2 = r = O(1),$$

and hence with Assumption (L1),

$$\|\mathbf{f}_{Z,t}\|^2 \leq \|\mathbf{Z}_{\otimes}^{1/2}\|_F^2 \cdot \|\mathbf{f}_t\|^2 = O_P\left(\prod_{j=1}^K d_j^{\alpha_{j,1}}\right) = O_P(g_s). \quad (\text{S.60})$$

With (S.57), (S.59) and the above result,

$$\begin{aligned} \|\tilde{\mathbf{f}}_{Z,t}\|^2 &= O_P(1) \cdot \left(\sum_{j=1}^d \|\hat{\mathbf{Q}}_{\otimes,j}\|^2\right) \left(\sum_{j=1}^d \|\mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} - \hat{\mathbf{Q}}_{\otimes,j}\|^2\right) \\ &\quad \cdot \|(\boldsymbol{\Sigma}_{A,K} \otimes \cdots \otimes \boldsymbol{\Sigma}_{A,1})^{-1}\|_F^2 \cdot \|\mathbf{H}_{\otimes}^{-1}\|_F^6 \cdot \|\mathbf{f}_{Z,t}\|^2 \\ &= O_P\left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1}-2\alpha_{k,r_k}} g_s^{-1} + d^2 g_s^{-1} d_k^{2\alpha_{k,1}-3\alpha_{k,r_k}-1} + g_{\eta} g_s d_k^{2\alpha_{k,1}-3\alpha_{k,r_k}+1} \right\}\right), \end{aligned} \quad (\text{S.61})$$

where we also used Assumption (L1) in the last equality. Similarly, by (S.56),

$$\begin{aligned} \|\tilde{\boldsymbol{\varepsilon}}_t\|^2 &= O_P(1) \cdot \|(\boldsymbol{\Sigma}_{A,K} \otimes \cdots \otimes \boldsymbol{\Sigma}_{A,1})^{-1}\|_F^2 \cdot \|\mathbf{H}_{\otimes}^{-1}\|_F^4 \cdot O_P\left(\max_{j \in [d]} \|\mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} - \hat{\mathbf{Q}}_{\otimes,j}\|^2\right) \left(\sum_{j,\ell=1}^d |\mathbb{E}\varepsilon_{t,j}\varepsilon_{t,\ell}|\right) \\ &= O_P\left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1}-\alpha_{k,r_k}} g_s^{-2} g_w^{-1} + d^3 g_s^{-2} g_w^{-1} d_k^{2\alpha_{k,1}-2\alpha_{k,r_k}-2} + d g_{\eta} g_w^{-1} d_k^{2\alpha_{k,1}-2\alpha_{k,r_k}} \right\}\right), \end{aligned} \quad (\text{S.62})$$

since $\sum_{j,\ell=1}^d |\mathbb{E}\varepsilon_{t,j}\varepsilon_{t,\ell}| = O(d)$ by Assumption (E1). By the same token,

$$\|\tilde{\boldsymbol{\varepsilon}}_{H,t}\|^2 = O_P(1) \cdot \|(\boldsymbol{\Sigma}_{A,K} \otimes \cdots \otimes \boldsymbol{\Sigma}_{A,1})^{-1}\|_F^2 \cdot \|\mathbf{H}_{\otimes}^{-1}\|_F^2 \cdot \|\mathbf{Z}_{\otimes}^{-1/2}\|_F^2 \cdot \left\| \sum_{j=1}^d m_{t,j} \mathbf{A}_{\otimes,j} \boldsymbol{\varepsilon}_{t,j} \right\|^2 = O_P(d/g_w), \quad (\text{S.63})$$

where we used

$$\mathbb{E} \left\| \sum_{j=1}^d m_{t,j} \mathbf{A}_{\otimes,j} \boldsymbol{\varepsilon}_{t,j} \right\|^2 \leq \max_{j \in [d]} \|\mathbf{A}_{\otimes,j}\|_F^2 \sum_{j,\ell=1}^d |\mathbb{E}\varepsilon_{t,j}\varepsilon_{t,\ell}| = O(d).$$

Therefore, we have from (S.61), (S.63) and (S.62),

$$\begin{aligned} \|\hat{\mathbf{f}}_{Z,t} - (\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t}\|^2 &\leq \|\tilde{\boldsymbol{\varepsilon}}_{H,t}\|^2 + \|\tilde{\boldsymbol{\varepsilon}}_t\|^2 + \|\tilde{\mathbf{f}}_{Z,t}\|^2 \\ &= O_P\left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1}-2\alpha_{k,r_k}} g_s^{-1} + d^2 g_s^{-1} d_k^{2\alpha_{k,1}-3\alpha_{k,r_k}-1} + g_{\eta} g_s d_k^{2\alpha_{k,1}-3\alpha_{k,r_k}+1} \right\} + \frac{d}{g_w}\right), \end{aligned}$$

where we also used $1/2 < \alpha_{k,r_k} \leq \alpha_{k,1} \leq 1$ from Assumption (L1) to conclude that $d g_s^{-1} g_w^{-1} = o(1)$, so that in fact $\|\tilde{\boldsymbol{\varepsilon}}_t\|^2 = o_P(\|\tilde{\mathbf{f}}_{Z,t}\|^2)$.

Now from (S.58) and using the notations in (S.51), we can obtain the vectorized imputed values, which are the vectorized estimated common components, as $\hat{\mathbf{c}}_t = \hat{\mathbf{Q}}_{\otimes} \hat{\mathbf{f}}_{Z,t}$ for any $t \in [T]$. Then for $j \in [d]$, we have the squared individual imputation error as

$$\begin{aligned} (\hat{\mathcal{C}}_{t,i_1,\dots,i_K} - \mathcal{C}_{t,i_1,\dots,i_K})^2 &= (\hat{\mathbf{c}}_t - \mathbf{c}_t)_j^2 = (\hat{\mathbf{Q}}'_{\otimes,j} \hat{\mathbf{f}}_{Z,t} - \mathbf{Q}'_{\otimes,j} \mathbf{f}_{Z,t})^2 \\ &= \left[\left(\hat{\mathbf{Q}}_{\otimes,j} - \mathbf{H}_{\otimes} \mathbf{Q}_{\otimes,j} \right)' \left((\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t} + \tilde{\boldsymbol{\varepsilon}}_{H,t} + \tilde{\boldsymbol{\varepsilon}}_t + \tilde{\mathbf{f}}_{Z,t} \right) + \mathbf{A}'_{\otimes,j} \mathbf{Z}_{\otimes}^{-1/2} \mathbf{H}'_{\otimes} \left(\tilde{\boldsymbol{\varepsilon}}_{H,t} + \tilde{\boldsymbol{\varepsilon}}_t + \tilde{\mathbf{f}}_{Z,t} \right) \right]^2 \\ &= O_P\left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1}-2\alpha_{k,r_k}} g_s^{-1} g_w^{-1} + d^2 g_s^{-1} g_w^{-1} d_k^{2\alpha_{k,1}-3\alpha_{k,r_k}-1} + g_{\eta} g_s g_w^{-1} d_k^{2\alpha_{k,1}-3\alpha_{k,r_k}+1} \right\} + \frac{d}{g_w^2}\right), \end{aligned}$$

where we used (S.56), (S.57) and Assumption (R1) in the last equality.

Lastly, we have the average imputation error as the following,

$$\begin{aligned}
& \frac{1}{Td} \sum_{t=1}^T \sum_{i_1, \dots, i_K=1}^{d_1, \dots, d_K} (\hat{\mathcal{C}}_{t, i_1, \dots, i_K} - \mathcal{C}_{t, i_1, \dots, i_K})^2 \\
&= \frac{1}{Td} \sum_{t=1}^T \|\hat{\mathbf{c}}_t - \mathbf{c}_t\|^2 = \frac{1}{Td} \sum_{t=1}^T \|\hat{\mathbf{Q}}_{\otimes} \hat{\mathbf{f}}_{Z,t} - \mathbf{Q}_{\otimes} \mathbf{f}_{Z,t}\|^2 \\
&= \frac{1}{Td} \sum_{t=1}^T \left\| (\hat{\mathbf{Q}}_{\otimes} - \mathbf{Q}_{\otimes} \mathbf{H}'_{\otimes}) (\mathbf{H}'_{\otimes})^{-1} \mathbf{f}_{Z,t} + \hat{\mathbf{Q}}_{\otimes} \left(\tilde{\boldsymbol{\varepsilon}}_{H,t} + \tilde{\boldsymbol{\varepsilon}}_t + \tilde{\mathbf{f}}_{Z,t} \right) \right\|^2 \\
&= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-1} + d g_s^{-1} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 1} + d^{-1} g_{\eta} g_s d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} + 1} \right\} + \frac{1}{g_w} \right), \tag{S.64}
\end{aligned}$$

where we used (S.61), (S.63), (S.62) and Lemma 6 in the last equality, and the fact that $\|\hat{\mathbf{Q}}_{\otimes}\|_F^2 = r = O(1)$.

This completes the proof of Theorem 2. $\square$

Proof of Theorem 5. Firstly, we use the notations in (S.2), and define also $\mathbf{Z} := \mathbf{Z}_k$ and $\mathbf{R}^* := \mathbf{R}_k^*$, which coincides with the $\mathbf{R}^*$ defined in (S.20). Then for $j \in [r_k]$,

$$\begin{aligned}
\lambda_j(\mathbf{R}^*) &= \lambda_j \left(\frac{1}{T} \sum_{t=1}^T \mathbf{Q} \mathbf{F}_{Z,t} \mathbf{A}' \mathbf{A} \mathbf{F}'_{Z,t} \mathbf{Q}' \right) = \lambda_j \left(\frac{1}{T} \sum_{t=1}^T \mathbf{A}_k \mathbf{F}_t \left[\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j \right]' \left[\otimes_{j \in [K] \setminus \{k\}} \mathbf{A}_j \right] \mathbf{F}'_t \mathbf{A}'_k \right) \\
&= \lambda_j \left(\mathbf{A}'_k \mathbf{A}_k \cdot \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \left[\otimes_{\ell \in [K] \setminus \{k\}} \mathbf{A}_{\ell} \right]' \left[\otimes_{\ell \in [K] \setminus \{k\}} \mathbf{A}_{\ell} \right] \mathbf{F}'_t \right) \\
&\asymp_P \lambda_j(\mathbf{A}'_k \mathbf{A}_k \cdot \text{tr}(\otimes_{\ell \in [K] \setminus \{k\}} \mathbf{A}'_{\ell} \mathbf{A}_{\ell})) = \lambda_j(\mathbf{A}'_k \mathbf{A}_k) \prod_{\ell \in [K] \setminus \{k\}} \text{tr}(\mathbf{A}'_{\ell} \mathbf{A}_{\ell}) \\
&\asymp \lambda_j(\mathbf{Z} \mathbf{Q}' \mathbf{Q}) \prod_{\ell \in [K] \setminus \{k\}} \sum_{i=1}^{r_k} d_{\ell}^{\alpha_{\ell,i}} \asymp \lambda_j(\boldsymbol{\Sigma}_{A,k}^{1/2} \mathbf{Z} \boldsymbol{\Sigma}_{A,k}^{1/2}) \cdot \prod_{\ell \in [K] \setminus \{k\}} d_{\ell}^{\alpha_{\ell,1}} \\
&\asymp \lambda_j(\mathbf{Z}) d_k^{-\alpha_{k,1}} g_s \asymp g_s d_k^{\alpha_{k,j} - \alpha_{k,1}}, \tag{S.65}
\end{aligned}$$

where the third line uses Assumption (F1), and Assumption (L1) in the second last line. The last line uses Theorem 1 of Ostrowski (1959) on the eigenvalues of a congruent transformation $\boldsymbol{\Sigma}_{A,k}^{1/2} \mathbf{Z} \boldsymbol{\Sigma}_{A,k}^{1/2}$ of $\mathbf{Z}$, and from Assumption (L1) that $\boldsymbol{\Sigma}_{A,k}$ has eigenvalues uniformly bounded away from 0 and infinity.

Since $\hat{\mathbf{S}} = \mathbf{R}^* + (\hat{\mathbf{S}} - \mathbf{R}^*)$, for $j \in [r_k]$, we have by Weyl's inequality that

$$\begin{aligned}
|\lambda_j(\hat{\mathbf{S}}) - \lambda_j(\mathbf{R}^*)| &\leq \|\hat{\mathbf{S}} - \mathbf{R}^*\| \leq \|\tilde{\mathbf{R}} - \mathbf{R}^*\| + \|\mathbf{R}_1\| + \|\mathbf{R}_2\| + \|\mathbf{R}_3\| \\
&\leq \omega_k \left\{ \sup_{\|\boldsymbol{\gamma}\|=1} |\tilde{R}(\boldsymbol{\gamma}) - R^*(\boldsymbol{\gamma})| + \sup_{\|\boldsymbol{\gamma}\|=1} R_1 + \sup_{\|\boldsymbol{\gamma}\|=1} R_2 + \sup_{\|\boldsymbol{\gamma}\|=1} R_3 \right\} = o_P(\omega_k), \tag{S.66}
\end{aligned}$$

where we use the decomposition in (S.20) in the first line, and $\omega_k := g_s d_k^{\alpha_{k,r_k} - \alpha_{k,1}}$ is defined at the beginning of the proof of Lemma 2. The second line uses $\tilde{R}(\boldsymbol{\gamma})$, $R^*(\boldsymbol{\gamma})$, R_1 , R_2 and R_3 defined in (S.21), and the convergence in probability in (S.25) and (S.30).

Secondly, with Assumption (R1) and our choice of ξ (see also (S.23)),

$$\xi / \omega_k \asymp d g_s^{-1} d_k^{\alpha_{k,1} - \alpha_{k,r_k}} [(T d_{-k})^{-1/2} + d_k^{-1/2}] = o(1). \tag{S.67}$$

For $r_k > 1$, if $j \in [r_k - 1]$, using (S.66) and (S.67), consider

$$\begin{aligned} \frac{\lambda_{j+1}(\widehat{\mathbf{S}}) + \xi}{\lambda_j(\widehat{\mathbf{S}}) + \xi} &\leq \frac{\lambda_{j+1}(\mathbf{R}^*) + \xi + |\lambda_{j+1}(\widehat{\mathbf{S}}) - \lambda_{j+1}(\mathbf{R}^*)|}{\lambda_j(\mathbf{R}^*) + \xi - |\lambda_j(\widehat{\mathbf{S}}) - \lambda_j(\mathbf{R}^*)|} = \frac{\lambda_{j+1}(\mathbf{R}^*) + o_P(\omega_k)}{\lambda_j(\mathbf{R}^*) + o_P(\omega_k)} \\ &= \frac{\lambda_{j+1}(\mathbf{R}^*)}{\lambda_j(\mathbf{R}^*)} (1 + o_P(1)) \asymp_P d_k^{\alpha_k, j+1 - \alpha_k, j}, \end{aligned} \quad (\text{S.68})$$

where the last line uses (S.65). Also, for $j \in [r_k - 1]$,

$$\frac{\lambda_{r_k+1}(\widehat{\mathbf{S}}) + \xi}{\lambda_{r_k}(\widehat{\mathbf{S}}) + \xi} = \frac{\lambda_{r_k+1}(\widehat{\mathbf{S}}) + \xi}{\omega_k(1 + o_P(1))} = O_P\left(\frac{\lambda_{r_k+1}(\widehat{\mathbf{S}})}{\omega_k} + \frac{\xi}{\omega_k}\right) \quad (\text{S.69})$$

$$\begin{aligned} &= O_P\left(\sup_{\|\gamma\|=1} (\tilde{R}(\gamma) - R^*(\gamma) + R_1 + R_2 + R_3) + \xi/\omega_k\right) \\ &= O_P(\xi/\omega_k) = o_P(d_k^{\alpha_k, j+1 - \alpha_k, j}), \end{aligned} \quad (\text{S.70})$$

where the second last equality uses (S.23), (S.24) and (S.30) together with our choice of ξ , and the last equality uses the extra rate assumption in the statement of the theorem. In the third equality, we assume the following is true (to be shown at the end of this proof):

$$\lambda_j(\widehat{\mathbf{S}}) = \lambda_j((\tilde{\mathbf{R}} - \mathbf{R}^*) + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3), \quad j = r_k + 1, \dots, d_k, \quad (\text{S.71})$$

so that

$$\frac{\lambda_j(\widehat{\mathbf{S}})}{\omega_k} = \lambda_j\left(\frac{1}{\omega_k}((\tilde{\mathbf{R}} - \mathbf{R}^*) + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3)\right) \leq \sup_{\|\gamma\|=1} ((\tilde{R}(\gamma) - R^*(\gamma)) + R_1 + R_2 + R_3).$$

Hence for $j = r_k + 1, \dots, \lfloor d_k/2 \rfloor$ (true also for $r_k = 1$),

$$\frac{\lambda_{j+1}(\widehat{\mathbf{S}}) + \xi}{\lambda_j(\widehat{\mathbf{S}}) + \xi} \geq \frac{\xi/\omega_k}{\sup_{\|\gamma\|=1} ((\tilde{R}(\gamma) - R^*(\gamma)) + R_1 + R_2 + R_3) + \xi/\omega_k} \geq \frac{1}{C} \quad (\text{S.72})$$

in probability for some generic constant $C > 0$, where the last inequality uses (S.23), (S.24) and (S.30) together with our choice of ξ . Combining (S.68), (S.70) and (S.72), we can easily see that our proposed $\widehat{r}_k$ is a consistent estimator for r_k .

If $r_k = 1$, then (S.70) becomes

$$\frac{\lambda_{r_k+1}(\widehat{\mathbf{S}}) + \xi}{\lambda_{r_k}(\widehat{\mathbf{S}}) + \xi} = O_P(\xi/\omega_k) = o_P(1).$$

When combined with (S.72) which is true also for $r_k = 1$, we can see that $\widehat{r}_k = 1$ in probability, showing that $\widehat{r}_k$ is a consistent estimator of r_k .

It remains to show (S.71). To this end, from (S.65) and (S.66), the first r_k eigenvalues of $\widehat{\mathbf{S}}$ coincides with those of $\mathbf{R}^*$ asymptotically, so that the first r_k eigenvectors corresponding to $\widehat{\mathbf{S}}$ coincides with those for $\mathbf{R}^*$ asymptotically as $T, d_k \rightarrow \infty$, which are necessarily in $\mathcal{N}^\perp := \text{Span}(\mathbf{Q})$, the linear span of the columns of $\mathbf{Q}$ (see (4.6), where $\mathbf{R}^*$ is sandwiched by $\mathbf{Q}$ and $\mathbf{Q}'$). This means that the $(r_k + 1)$ -th largest eigenvalue of $\widehat{\mathbf{S}}$ and beyond will asymptotically have eigenvectors in $\mathcal{N}$, the orthogonal complement of $\mathcal{N}^\perp$. Then for any unit vectors $\gamma \in \mathcal{N}$, we have from the definitions of $\mathbf{R}^*$, $\tilde{\mathbf{R}}$, $\mathbf{R}_1$, $\mathbf{R}_2$ and $\mathbf{R}_3$ in (S.21) that

$$\gamma' \widehat{\mathbf{S}} \gamma = \gamma' (\mathbf{R}^* + (\tilde{\mathbf{R}} - \mathbf{R}^*) + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3) \gamma = \gamma' ((\tilde{\mathbf{R}} - \mathbf{R}^*) + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3) \gamma,$$

which is equivalent to (S.71). This completes the proof of the theorem. $\square$

Proof of Proposition 1. We can show stable convergence in law similar to Proposition 3.1 in Xiong and Pelger (2023). First, using Assumption (F1) we can write

$$\begin{aligned} & \sqrt{T d_k^{\alpha_k, r_k}} \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \sum_{i=1}^{d_k} \mathbf{Q}_{k,i} \mathbf{A}'_{k,i} \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{A}_{k,j} \\ &= \sum_{t=1}^T \sqrt{\frac{d_k^{\alpha_k, r_k}}{T}} \cdot \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij,h}\}}{|\psi_{k,ij,h}|} - 1 \right) \\ & \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \mathbf{Q}_{k,i} \mathbf{A}'_{k,i} (\text{mat}_k(\mathcal{F}_t) \mathbf{v}_{k,h} \mathbf{v}'_{k,h} \text{mat}_k(\mathcal{F}_t)' - \mathbf{v}'_{k,h} \mathbf{v}_{k,h} \Sigma_k) \mathbf{A}_{k,j}. \end{aligned}$$

Define the filtration $\mathcal{G}^T := \sigma(\cup_{s=1}^T \mathcal{G}_s)$ where the sigma-algebra $\mathcal{G}_s := \sigma(\{\mathcal{M}_{t,i_1,\dots,i_K} \mid t \leq s\}, \mathbf{A}_1, \dots, \mathbf{A}_K)$. Let $\mathbf{u} \in \mathbb{R}^{r_k}$ be a non-random unit vector. For a given $k \in [K], j \in [d_k]$, define also the random variable

$$\begin{aligned} g_{k,j,t} &:= \mathbf{u}' \sqrt{\frac{d_k^{\alpha_k, r_k}}{T}} \cdot \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij,h}\}}{|\psi_{k,ij,h}|} - 1 \right) \\ & \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} \mathbf{Q}_{k,i} \mathbf{A}'_{k,i} (\text{mat}_k(\mathcal{F}_t) \mathbf{v}_{k,h} \mathbf{v}'_{k,h} \text{mat}_k(\mathcal{F}_t)' - \mathbf{v}'_{k,h} \mathbf{v}_{k,h} \Sigma_k) \mathbf{A}_{k,j}. \end{aligned}$$

Since each entry in $\mathcal{F}_t$ is i.i.d. by Assumption (F1) and is independent of $(\mathcal{M}_t, \mathbf{A}_1, \dots, \mathbf{A}_K)$ by Assumptions (O1) and (L1), we have $\mathbb{E}[g_{k,j,t} \mid \mathcal{G}_{t-1}] = 0$. Define $\Xi_{F,k} := \text{var}[\text{vec}(\text{mat}_k(\mathcal{F}_t) \mathbf{A}'_{-k} \mathbf{A}_{-k} \text{mat}_k(\mathcal{F}_t)' - \text{tr}(\mathbf{A}_{-k} \mathbf{A}'_{-k}) \Sigma_k)]$ and $\mathbf{x}_{F,k,j,il} := \text{vec}([\mathbf{A}'_{k,j} \otimes (\mathbf{D}^{-1} \mathbf{H}_k^{a,*} \mathbf{Q}_{k,i} \mathbf{A}'_{k,i})] \Xi_{F,k} [\mathbf{A}'_{k,j} \otimes (\mathbf{D}^{-1} \mathbf{H}_k^{a,*} \mathbf{Q}_{k,l} \mathbf{A}'_{k,l})]')$, so that we have

$$\|\mathbf{x}_{F,k,j,il}\|^2 \leq \|\Xi_{F,k}\|_F^2 \cdot \|\mathbf{D}^{-1}\|_F^4 \cdot \|\mathbf{Z}_k^{-1/2}\|_F^4 = O_P\left(d_k^{4\alpha_k,1-6\alpha_k,r_k} \prod_{j=1}^K d_j^{-4\alpha_{j,1}} \prod_{j \in [K] \setminus \{k\}} d_j^{4\alpha_{j,1}}\right) = O_P(d_k^{-6\alpha_k,r_k}),$$

leading to

$$\begin{aligned} & d_k^{\alpha_k, r_k} \cdot \mathbb{E} \left\| \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij}\}}{|\psi_{k,ij}|} - 1 \right) \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,lj}|} - 1 \right) (\mathbf{x}_{F,k,j,il} - \mathbb{E}[\mathbf{x}_{F,k,j,il}]) \right\|^2 \\ &= O_P(d_k^{4-5\alpha_k,r_k}) = o_P(1). \end{aligned}$$

Hence, it holds that

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}[g_{k,j,t}^2 \mid \mathcal{G}_{t-1}] &= \frac{d_k^{\alpha_k, r_k}}{T} \cdot \sum_{t=1}^T \mathbb{E} \left\{ \mathbf{u}' \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij}\}}{|\psi_{k,ij}|} - 1 \right) \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,lj}|} - 1 \right) \right. \\ & \cdot \mathbf{D}^{-1} \mathbf{H}_k^{a,*} [\mathbf{A}'_{k,j} \otimes (\mathbf{Q}_{k,i} \mathbf{A}'_{k,i})] \text{vec}(\text{mat}_k(\mathcal{F}_t) \mathbf{A}'_{-k} \mathbf{A}_{-k} \text{mat}_k(\mathcal{F}_t)' - \text{tr}(\mathbf{A}_{-k} \mathbf{A}'_{-k}) \Sigma_k) \\ & \cdot \text{vec}(\text{mat}_k(\mathcal{F}_t) \mathbf{A}'_{-k} \mathbf{A}_{-k} \text{mat}_k(\mathcal{F}_t)' - \text{tr}(\mathbf{A}_{-k} \mathbf{A}'_{-k}) \Sigma_k)' \\ & \left. \cdot [\mathbf{A}'_{k,j} \otimes (\mathbf{Q}_{k,l} \mathbf{A}'_{k,l})]' (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1} \mathbf{u} \mid \mathcal{G}_{t-1} \right\} \\ & \xrightarrow{p} d_k^{2+\alpha_k, r_k} \cdot \lim_{d_k \rightarrow \infty} \frac{1}{d_k^2} \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij}\}}{|\psi_{k,ij}|} - 1 \right) \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,lj}|} - 1 \right) \\ & \cdot \mathbf{u}' \mathbf{D}^{-1} \mathbf{H}_k^{a,*} [\mathbf{A}'_{k,j} \otimes (\mathbf{Q}'_k \mathbf{A}_k)] \Xi_{F,k} [\mathbf{A}'_{k,j} \otimes (\mathbf{Q}'_k \mathbf{A}_k)]' (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1} \mathbf{u} \\ & \xrightarrow{p} d_k^{2+\alpha_k, r_k} \omega_{\psi,k,j} \cdot \mathbf{u}' \mathbf{D}^{-1} \mathbf{H}_k^{a,*} [\mathbf{A}'_{k,j} \otimes (\mathbf{Q}'_k \mathbf{A}_k)] \Xi_{F,k} [\mathbf{A}'_{k,j} \otimes (\mathbf{Q}'_k \mathbf{A}_k)]' (\mathbf{H}_k^{a,*})' \mathbf{D}^{-1} \mathbf{u}, \end{aligned}$$

which satisfies the nesting condition of Theorem 6.1 in Häusler and Luschgy (2015). From Assumption (O1), we have $|(T \cdot \mathbb{1}\{t \in \psi_{k,ij,h}\})/|\psi_{k,ij,h}| - 1| \leq \max(\psi_0^{-1} - 1, 1)$. Hence with ϵ from Proposition 1, we

have

$$\begin{aligned}
& \sum_{t=1}^T \mathbb{E}[g_{k,j,t}^{2+\epsilon} \mid \mathcal{G}_{t-1}] \leq \|\mathbf{A}_{k,j\cdot}\|^{2+\epsilon} \cdot \frac{d_k^{\alpha_{k,r_k}(1+\epsilon/2)}}{T^{1+\epsilon/2}} \cdot \|\mathbf{D}^{-1} \mathbf{H}_k^{a,*}\|^{2+\epsilon} \\
& \cdot \sum_{t=1}^T \left\| \sum_{i=1}^{d_k} \sum_{h=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij,h}\}}{|\psi_{k,ij,h}|} - 1 \right) \mathbf{Q}_{k,i} \mathbf{A}'_{k,i} (\text{mat}_k(\mathcal{F}_t) \mathbf{v}_{k,h} \mathbf{v}'_{k,h} \text{mat}_k(\mathcal{F}_t)' - \mathbf{v}'_{k,h} \mathbf{v}_{k,h} \boldsymbol{\Sigma}_k) \right\|^{2+\epsilon} \\
& = O_P(1) \cdot (d_k^{\alpha_{k,1}-\alpha_{k,r_k}} g_s^{-1})^{2+\epsilon} \cdot \frac{d_k^{\alpha_{k,r_k}(1+\epsilon/2)}}{T^{\epsilon/2}} \cdot d_k^{-\alpha_{k,r_k}(1+\epsilon/2)} g_s^{2+\epsilon} = O_P\left(\frac{d_k^{\alpha_{k,1}-\alpha_{k,r_k}}}{T^{\epsilon/2}}\right) = o_P(1),
\end{aligned}$$

which is sufficient for the conditional Lindeberg condition in Häusler and Luschgy (2015) to hold. Then by the stable martingale central limit theorem (Theorem 6.1 in Häusler and Luschgy (2015)), we have

$$\sum_{t=1}^T g_{k,j,t} \rightarrow \mathcal{N}(0, \mathbf{D}^{-1} \mathbf{H}_k^{a,*} h_{k,j}(\mathbf{A}_{k,j\cdot})(\mathbf{H}_k^{a,*})' \mathbf{D}^{-1}) \quad \mathcal{G}^T\text{-stably as } T \rightarrow \infty,$$

where $h_{k,j}(\mathbf{A}_{k,j\cdot}) = d_k^{2+\alpha_{k,r_k}} \omega_{\psi,k,j} \cdot [\mathbf{A}'_{k,j\cdot} \otimes (\mathbf{Q}'_k \mathbf{A}_k)] \boldsymbol{\Xi}_{F,k} [\mathbf{A}'_{k,j\cdot} \otimes (\mathbf{Q}'_k \mathbf{A}_k)]'$. This completes the proof of Proposition 1. $\square$

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